

Texts in Applied Mathematics

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Chapter 6

Harvesting in Two-species Models

Cosecha de esfuerzo constante: Efecto proporcional al tamaño poblacional; mueve isoclinas paralelamente.

Cosecha de rendimiento constante: Efecto independiente del tamaño; crea hipérbolas; puede llevar a extinción en tiempo finito.

Tasa crítica : Máximo rendimiento para el cual existe equilibrio de coexistencia.

Tasa crítica verdadera : Puede ser menor que la tasa crítica en el "bad case".

Control bang-bang: Alterna entre extremos; óptimo para problemas lineales en el control.

Control singular: Ocurre cuando la función de conmutación se anula en un intervalo.

Equilibrio bionómico: Renta económica cero en pesquerías de acceso abierto.

Precio sombra: Valor marginal del stock; interpretación económica de la variable adjunta.

Cosecha intermitente: Puede lograr rendimientos cercanos a H_c incluso en casos donde la cosecha constante lleva al colapso.

Descuento: Afecta el equilibrio óptimo; tasas altas favorecen la explotación rápida.

6.1 Harvesting of Species in Competition

The topics in this chapter are part of the subject of natural resource management and bioeconomics. This is an important subject that is developing rapidly. The classical reference is the book by Clark (1990), where additional references may be found.

In Section 1.5 we examined the effects of both constant-yield harvesting and constant-effort harvesting on single-species populations modeled by an ordinary differential equation. In this section we shall study the effects of harvesting one of a pair of interacting species in competition.

In studying models for competition between two species (Section 5.1), we began by assuming linear per capita growth rates, obtaining a system

$$\begin{aligned}x' &= x(\lambda - ax - by), \\y' &= y(\mu - cx - dy).\end{aligned}\tag{6.1}$$

We observed that there are four cases depending on the relationship between the x -isocline $ax + by = \lambda$, on which $x' = 0$ and the y -isocline $cx + dy = \mu$, on which $y' = 0$. These cases may be depicted graphically in Figure 6.1, with equilibria marked $\times$ and asymptotically stable equilibria marked $\bullet$.

We will consider only the harvesting of one of the two species, say the x species. For constant-effort harvesting, the model would be

$$\begin{aligned}x' &= x(\lambda - ax - by) - Ex, \\y' &= y(\mu - cx - dy).\end{aligned}$$

This is the same as the unharvested model but with λ replaced by $\lambda - E$, or with the x -isocline moved parallel to itself toward the origin. It has boundary equilibria $(K(E), 0)$ and $(0, M)$ with $M = \frac{\mu}{d}$ and $K(E) = \frac{\lambda - E}{a + E}$. Examination of the four cases shows that it is possible by harvesting to change Case 1 to Case 4 (coexistence to x -extinction), Case 2 to Case 4 (survival of either species alone to x -extinction), or Case 3 to Case 1 or Case 2 and then to Case 4 (y -extinction to x -extinction via

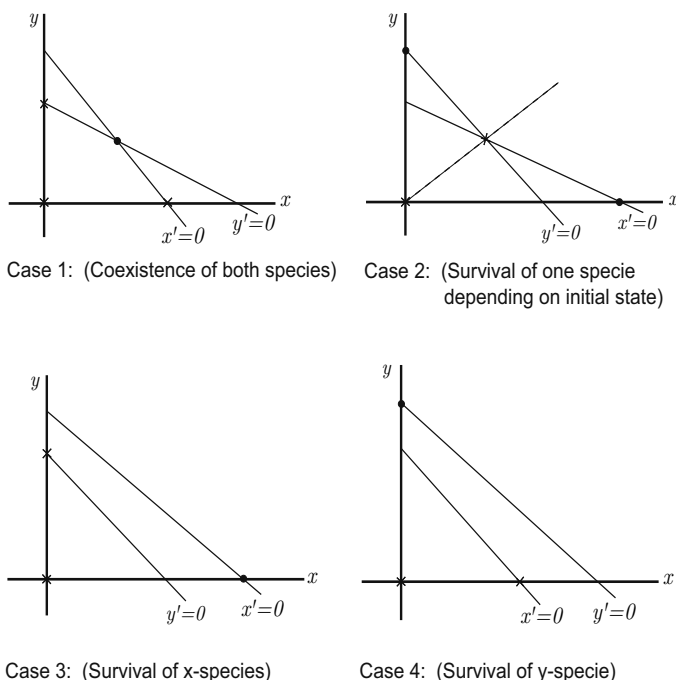


Fig. 6.1 Species in competition cases.

coexistence). If the growth rates are not linear, the same type of argument can be used to see how the isoclines move under harvesting and to obtain the same results, all of which are in accordance with our intuition.

With constant-yield harvesting, the model is

$$x' = x(\lambda - ax - by) - H, y' = y(\mu - cx - dy).$$

The x -isocline, instead of being the pair of lines $x = 0$, $ax + by = \lambda$, is now the curve $x(\lambda - ax - by) = H$, which is a hyperbola having the lines $x = 0$ and $ax + by = \lambda$ as asymptotes and which moves away from these asymptotes as H increases. The effect of harvesting on the system can be depicted graphically by sketching these hyperbolas; the results are somewhat different in the four cases.

Case 1, the asymptotically stable (coexistence) equilibrium moves up and to the left and the saddle point moves from $(0, M)$ down and to the right (see Figure 6.2). The stable separatrices at this saddle point move from the y -axis into the first quadrant, and divide this quadrant into a coexistence region (initial states that tend to the asymptotically stable equilibrium) and a region for which the x -species becomes extinct in finite time. Since $x = 0$ is not a solution of the harvested system, it is possible for orbits to cross the y -axis, corresponding to extinction of the x species in finite time. As the hyperbola moves down, the saddle point and asymptotically stable equilibrium coalesce and disappear, producing a catastrophe. For harvest rates

larger than the critical rate at which this coalescence occurs, the x species becomes extinct in finite time for all initial states.

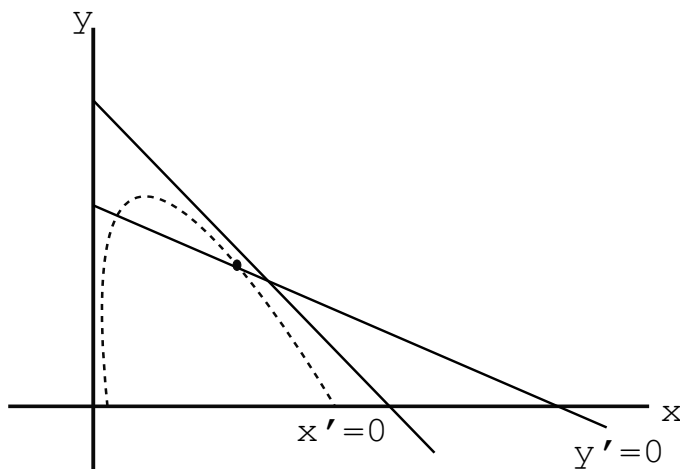


Fig. 6.2 Case 1.

Case 2, the asymptotically stable equilibrium at $(K, 0)$ moves to the left along the x -axis and the asymptotically stable equilibrium at $(0, M)$ moves into the second quadrant, corresponding to extinction of the x species in finite time (see Figure 6.3). The saddle point at (x_∞, y_∞) moves down and to the right, remaining a saddle point, and its stable separatrices divide the first quadrant into a region of x -extinction in finite time and a region of ultimate y -extinction. As the hyperbola moves down with increasing H , the equilibrium at (x_∞, y_∞) coalesces either with the equilibrium at $(0, 0)$ or the equilibrium at $(K, 0)$ and disappears; then the two remaining equilibria on the x -axis coalesce and disappear. The second coalescence signifies x -extinction for all initial states.

Case 3, the asymptotically stable equilibrium moves from $(K, 0)$ along the x -axis. If $ad - bc > 0$, when the hyperbola meets the line $cx + dy = \mu$ this equilibrium moves into the interior of the first quadrant, giving coexistence (see Figure 6.4(A)). The other equilibrium is a saddle point whose stable separatrices give a division into a coexistence region and a region of extinction of the x -species in finite time (see Figure 6.4(B)). When the two equilibria coalesce and disappear, we have extinction of the x -species in finite time for all initial states.

Case 4, the asymptotically stable equilibrium moves from $(0, M)$ into the second quadrant, giving extinction of the x species in finite time for all initial states (see Figure 6.5).

If we do not assume that the per capita growth rates are linear, but instead require only a model of Kolmogorov type,

$$x' = xf(x, y), \quad y' = yg(x, y),$$

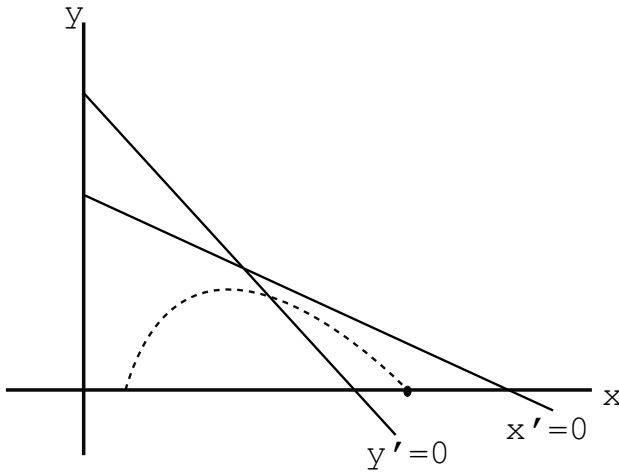


Fig. 6.3 Case 2.

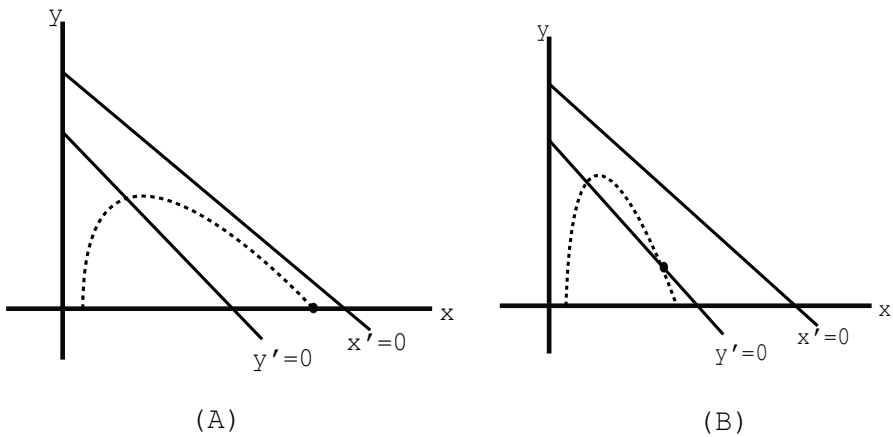


Fig. 6.4 Case 3.

with $f_x(x, y) < 0$, $f_y(x, y) < 0$, $g_x(x, y) < 0$, $g_y(x, y) < 0$ for $x > 0, y > 0$, then the isoclines are curves with negative slope but are not necessarily straight lines. The x -isocline under constant-yield harvesting is no longer necessarily a hyperbola, but it does approach the line $x = 0$ and the curve $f(x, y) = 0$ asymptotically. The same cases as those described for linear growth rates may occur, and the qualitative responses to harvesting are the same.

Example 1: Determine the response of the system

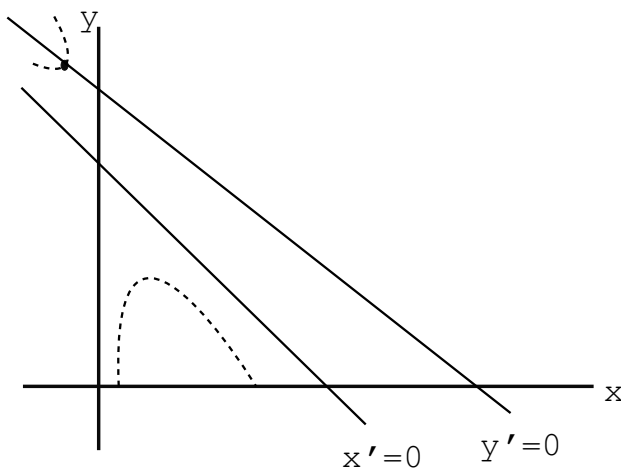


Fig. 6.5 Case 4.

$$\begin{aligned}x' &= x(100 - 4x - y), \\y' &= y(60 - x - 2y),\end{aligned}$$

to constant-effort harvesting of the x species.

Solution: With no harvesting, there is an asymptotically stable equilibrium at $(20, 20)$ [Example 1, Section 5.1]. A harvested system has the form

$$\begin{aligned}x' &= x[(100 - E) - 4x - y], \\y' &= y(60 - x - 2y).\end{aligned}$$

Equilibria are given by the pair of equations

$$4x + y = 100 - E, \quad x + 2y = 60,$$

and elimination gives the solution

$$x = 20 - \frac{2}{7}E, \quad y = 20 + \frac{E}{7}.$$

When $E = 70$, x reaches zero, and $y = 30$. Thus harvesting decreases the x -population size and increases the y -population size, eventually moving coexistence to x -extinction.

Example 2: Determine the response of the system

$$\begin{aligned}x' &= x(100 - 4x - y), \\y' &= y(60 - x - 2y),\end{aligned}$$

to constant-yield harvesting of the x species.

Solution: With no harvesting, there is an asymptotically stable equilibrium at $(20, 20)$ [Example 1, Section 5.1]. A harvested system has the form

$$\begin{aligned}x' &= x(100 - 4x - y) - H, \\y' &= y(60 - x - 2y).\end{aligned}$$

Equilibria are given by the pair of equations

$$x(100 - 4x - y) = H, \quad x + 2y = 60.$$

Replacing x by $60 - 2y$ in the first of these equations, we obtain a quadratic equation for y

$$\begin{aligned}(60 - 2y)[100 - 4(60 - 2y) - 2y] &= H, \\14(y - 30)(y - 20) + H &= 0, \\14y^2 - 700y + (8400 + H) &= 0.\end{aligned}$$

The roots of this equation may be found using the quadratic formula:

$$y = 25 \pm \frac{1}{2} \sqrt{100 - \frac{2H}{7}}.$$

For $H = 0$, these roots are $y = 20$ (which gives $x = 20$) and $y = 30$ (which gives $x = 0$). There is an asymptotically stable equilibrium at $(20, 20)$ and a saddle point at $(0, 30)$. As H increases, these equilibria move along the line $x + 2y = 60$ until they coalesce at $(10, 25)$ for $H = 350$.

Exercises

In each of Exercises 1–4, determine the response of the system to constant-effort harvesting of the x species.

1. $x' = x(80 - x - y)$, $y' = y(120 - x - 3y)$ [cf. Exercise 1, Section 5.1].
2. $x' = x(60 - 3x - y)$, $y' = y(75 - 4x - y)$ [cf. Exercise 2, Section 5.1].
3. $x' = x(40 - x - y)$, $y' = y(90 - x - 2y)$ [cf. Exercise 3, Section 5.1].
4. $x' = x(80 - 3x - 2y)$, $y' = y(80 - x - y)$ [cf. Exercise 4, Section 5.1].

In each of Exercises 5–8, determine the response of the system to constant-yield harvesting of the x species.

5. $x' = x(80 - x - y)$, $y' = y(120 - x - 3y)$ [cf. Exercise 1, Section 5.1].
6. $x' = x(60 - 3x - y)$, $y' = y(75 - 4x - y)$ [cf. Exercise 2, Section 5.1].
7. $x' = x(40 - x - y)$, $y' = y(90 - x - 2y)$ [cf. Exercise 3, Section 5.1].
8. $x' = x(80 - 3x - 2y)$, $y' = y(80 - x - y)$ [cf. Exercise 4, Section 5.1].

6.2 Harvesting of Predator–Prey Systems

Predator–prey systems are most commonly used to describe the interaction of a species with its food supply. In other words, the primary interest is in the predator population size, and the prey population size is of interest only because of its effect on the predator growth rate. A predator–prey model may be viewed as a refinement of a single-species model through more realistic description of the limitations on the growth rate imposed by resource limitations. Thus we will study only the harvesting of predators. It would be possible, however, to use an analogous approach to study the harvesting of prey and this would be appropriate in an examination of the extent to which one can control a population by tampering with its food supply.

Constant-effort harvesting: For a system modeled by

$$x' = xf(x) - xy\phi(x), \quad y' = y(cx\phi(x) - e),$$

(Rosenzweig–MacArthur model), the harvested system is modeled by

$$x' = xf(x) - yx\phi(x), \quad y' = y(cx\phi(x) - e - E).$$

The effect of harvesting is to move the predator isocline $x = J$, where $cJ\phi(J) = e$, to the right. This applies to the more general (Kolmogorov type) model

$$x' = xf(x, y), \quad y' = y(g(x) - E),$$

with the predator per capita growth rate independent of predator population size. If the prey isocline $y = \frac{f(x)}{\phi(x)}$ (as in the Rosenzweig–MacArthur model) or $f(x, y) = 0$ (as in the Kolmogorov model) is concave downwards and has a maximum, an equilibrium that is unstable for $E = 0$ is stabilized by harvesting as the harvested predator isocline moves past the maximum of the prey isocline (see Figure 6.6).

For a given model, the effort that maximizes yield may be calculated by solving

$$\frac{d}{dE}(Ey_\infty) = E \frac{dy_\infty}{dE} + y_\infty = 0$$

for E and then calculating $Ey_\infty(E)$ for this value of E . Alternatively, one would proceed as follows for a Kolmogorov model. For a given effort E , the equilibrium conditions are $f(x_\infty(E), y_\infty(E)) = 0$, $g(x_\infty(E)) = E$, and the yield is $Ey_\infty(E) = y_\infty(E)g(x_\infty(E))$. The maximum yield is thus the maximum of the function $yg(x)$ subject to the constraint $f(x, y) = 0$. So long as $E < cK\phi(K) - e$, where $f(K) = 0$ (Rosenzweig–MacArthur model) or $E < g(K)$, where $f(K, 0) = 0$ (Kolmogorov model), $y_\infty(E) > 0$ and the equilibrium depends continuously on E . For $E \geq cK\phi(K) - e$ or $E \geq g(K)$ in the respective models, we have $y_\infty(E) = 0$, corresponding to the ultimate extinction of predators.

Constant-yield harvesting: For a Rosenzweig–MacArthur model the harvested system is

$$x' = xf(x) - xy\phi(x), \quad y' = y(cx\phi(x) - e) - H.$$

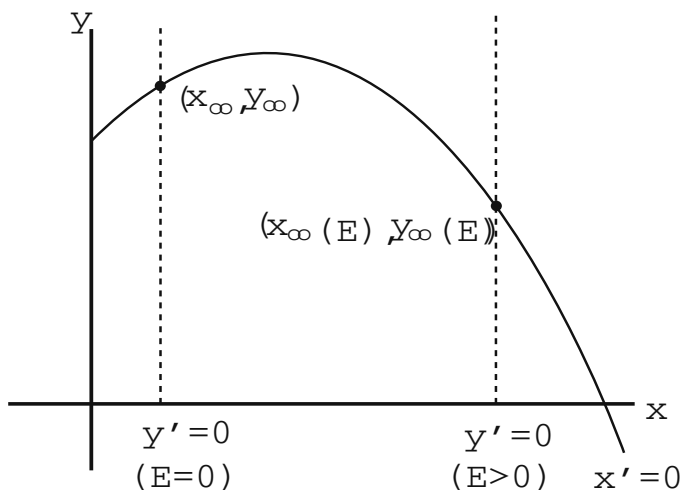


Fig. 6.6 Stabilization under harvesting.

The predator isocline $cx\phi(x) = e$ is replaced by the curve $y(cx\phi(x) - e) = H$ to the right of $cx\phi(x) = e$ and having $cx\phi(x) = e$ as a vertical asymptote (since $cx\phi(x) - e \rightarrow 0$ as $y \rightarrow \infty$) and a horizontal asymptote $y = \frac{H}{c \lim_{x \rightarrow \infty} x\phi(x) - e}$ (since $y \rightarrow \frac{H}{c \lim_{x \rightarrow \infty} x\phi(x) - e}$ as $x \rightarrow \infty$). It is no more difficult to study the more general system

$$x' = xf(x, y), \quad y' = yg(x),$$

or indeed the general Kolmogorov model

$$x' = xf(x, y), \quad y' = yg(x, y),$$

with harvested system

$$x' = xf(x, y), \quad y' = yg(x, y) - H,$$

in which $f_y(x, y) < 0$, $g_x(x, y) > 0$, $g_y(x, y) \leq 0$, $f(K, 0) = 0$, $g(J, 0) = 0$; we will usually think in terms of the special case where g is independent of y . As H increases, the “hyperbola-like” curve $y(cx\phi(x) - e) = H$ or $yg(x, y) = H$ moves away from the line $x = J$ or the curve $g(x, y) = 0$ and upward to the right.

It intersects the prey isocline $f(x, y) = 0$ at two points: $(x_\infty(H), y_\infty(H))$ coming from (x_∞, y_∞) when $H = 0$, and $(\xi(H), y(H))$ coming from $(K, 0)$ when $H = 0$ (Figure 6.7). As H increases, these two equilibria move together, coalesce, and disappear. The value of H for which they coalesce is the value for which the level curve $yg(x, y) = H$ of the function $yg(x, y)$ is tangent to the prey isocline. This is the largest value of the function $yg(x, y)$ attained on the curve $f(x, y) = 0$; we define

$$H_c = \max_{f(x, y)=0} yg(x, y).$$

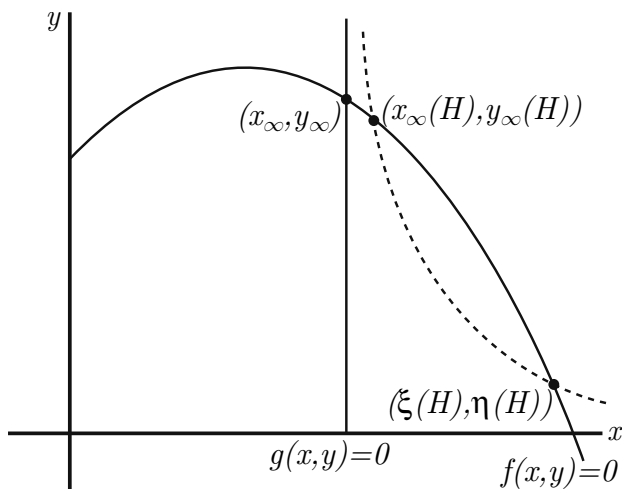


Fig. 6.7 Two equilibria.

Then for $H > H_c$ there are no equilibria in the first quadrant (Figure 6.8).

If $H > 0$, then $y' < 0$ on the x -axis. This means that solutions can reach and cross the x -axis in finite time (predator extinction). Indeed, if $H > H_c$, this occurs for all solutions, independent of initial values.

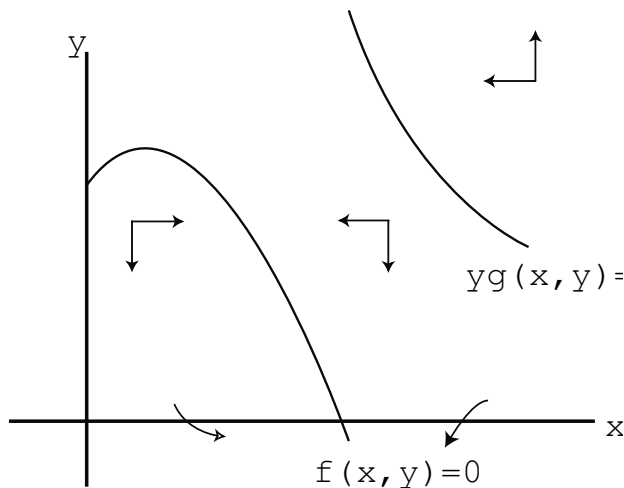


Fig. 6.8 No equilibrium.

It is possible to prove that no solution can be unbounded; the proof is similar to the proof in the case $H = 0$ but the details are more complicated. Thus a solution either tends to an equilibrium, or tends to a limit cycle, or reaches the x -axis in finite

time. If a solution reaches the x -axis in finite time, we consider the system to have collapsed.

To study the stability of any equilibrium (x_0, y_0) we look at the community matrix

$$A(x_0, y_0) = \begin{pmatrix} x_0 f_x(x_0, y_0) & x_0 f_y(x_0, y_0) \\ y_0 g_x(x_0, y_0) & g(x_0, y_0) + y_0 g_y(x_0, y_0) \end{pmatrix}.$$

If $\det A(x_0, y_0) < 0$, then (x_0, y_0) is a saddle point and if $\det A(x_0, y_0) > 0$, then (x_0, y_0) is a node or spiral point that is unstable if $\operatorname{tr} A(x_0, y_0) > 0$ and asymptotically stable if $\operatorname{tr} A(x_0, y_0) < 0$.

The slope of the prey isocline $f(x, y) = 0$ at (x_0, y_0) is $-\frac{f_x(x_0, y_0)}{f_y(x_0, y_0)}$, and the slope of the (harvested) predator isocline $yg(x, y) = H$ at (x_0, y_0) is $-\frac{y_0 g_x(x_0, y_0)}{y_0 g_y(x_0, y_0) + g(x_0, y_0)}$. If we assume $g(x_0, y_0) + y_0 g_y(x_0, y_0) \geq 0$, which is certainly true if $g(x, y)$ is independent of y , and use the hypothesis $f_y(x_0, y_0) < 0$, we see that the slope of the prey isocline is less than the slope of the predator isocline if and only if $f_x(yg_y + g) < yg_x f_y$, or

$$y_0(f_x(x_0, y_0)g_y(x_0, y_0) - f_y(x_0, y_0)g_x(x_0, y_0)) + g(x_0, y_0)f_x(x_0, y_0) < 0.$$

Because

$$\det A(x_0, y_0) = x_0 y_0 (f_x(x_0, y_0)g_y(x_0, y_0) - f_y(x_0, y_0)g_x(x_0, y_0)) + g(x_0, y_0)f_x(x_0, y_0),$$

(x_0, y_0) is a saddle point if and only if the slope of the prey isocline at (x_0, y_0) is less than the slope of the predator isocline at (x_0, y_0) . Thus $(\xi(H), y(H))$ is a saddle point and $(x_\infty(H), y_\infty(H))$ is not a saddle point. If $(x_\infty(H), y_\infty(H))$ is asymptotically stable, some orbits tend to it, and if $(x_\infty(H), y_\infty(H))$ is unstable, then either there is a limit cycle to which some orbits tend, or every orbit reaches the x -axis in finite time. We define the region of coexistence to be the set of initial population sizes $(x(0), y(0))$ for which the orbit tends either to the equilibrium $(x_\infty(H), y_\infty(H))$ or to a limit cycle around this equilibrium as $E \rightarrow \infty$. We know that if $H = 0$, the region of coexistence is the whole first quadrant of the x - y plane. If $H > 0$, the region of coexistence is smaller and depends on the separating orbits at the saddle point $(\xi(H), \eta(H))$. There is one stable separatrix in the region $x' < 0, y' > 0$, and there is one unstable separatrix in the region $x' > 0, y' < 0$ (Figure 6.9).

There are two possible cases for the other two separatrices:

“Good case”: There is a stable separatrix running from $x \rightarrow +\infty$ around $(x_\infty(H), y_\infty(H))$ to $(\xi(H), \eta(H))$ and an unstable separatrix running from $(\xi(H), \eta(H))$ to $(x_\infty(H), y_\infty(H))$ or a limit cycle around $(x_\infty(H), y_\infty(H))$. The coexistence region is bounded by the two stable separatrices, and orbits starting in this region tend to $(x_\infty(H), y_\infty(H))$ or to a limit cycle around $(x_\infty(H), y_\infty(H))$ (Figure 6.10).

“Bad case”: There is a stable separatrix running from the equilibrium $(x_\infty(H), y_\infty(H))$ or from a periodic solution around $(x_\infty(H), y_\infty(H))$ and an unstable separatrix around $(x_\infty(H), y_\infty(H))$ and then down to the x -axis. Either there is no coexistence region at all, or there is an unstable periodic orbit around the asymptotically sta-

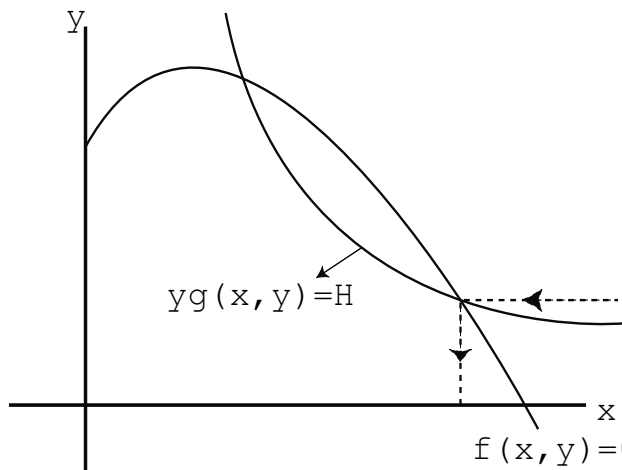


Fig. 6.9 Stable and unstable separatrices.

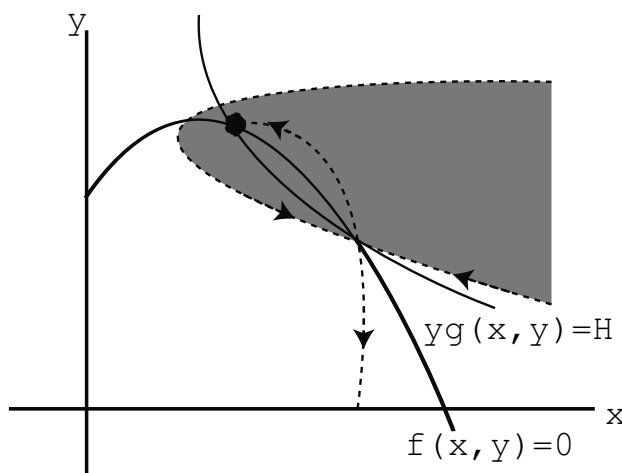


Fig. 6.10 “Good case”.

ble equilibrium $(x_\infty(H), y_\infty(H))$ and the coexistence region is the interior of this periodic orbit (Figure 6.11).

It is possible to have a transition from the “good” to the “bad” case as H increases. If this occurs, then at the transition there must be a homoclinic orbit from saddle point to saddle point (Figure 6.12).

For $H = 0$, the system must be in the “good” case since the unstable separatrix at $(K, 0)$ cannot return to the x -axis and must therefore go to (x_∞, y_∞) or to a limit cycle around (x_∞, y_∞) . Since the orbits depend continuously on the system, the system must be in the “good” case if H is sufficiently small.

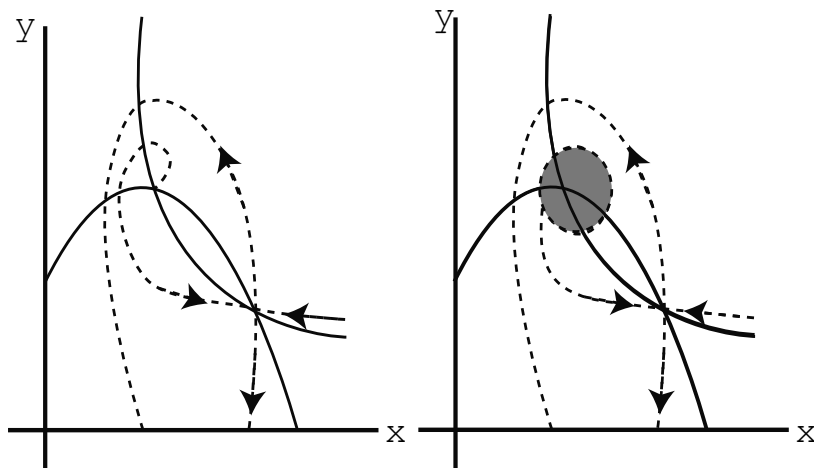


Fig. 6.11 “Bad case”.

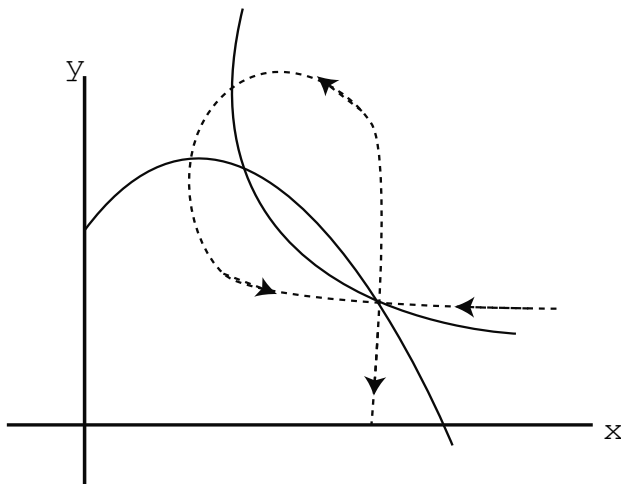


Fig. 6.12 Homoclinic orbit.

For H close to H_c we can decide whether the system is in the “good” or “bad” case by looking at the local stability of the equilibrium $(x_\infty(H), y_\infty(H))$, as is shown by the following theorem.

Theorem 6.1. Suppose that $\text{tr}A(x_\infty(H_c), y_\infty(H_c)) \neq 0$. Then for $H < H_c$ but H sufficiently close to H_c , the system is in the “good” case if $(x_\infty(H_c), y_\infty(H_c))$ is asymptotically stable, that is, if $\text{tr}A(x_\infty(H_c), y_\infty(H_c)) < 0$, and it is in the “bad” case if $(x_\infty(H_c), y_\infty(H_c))$ is unstable, that is, if $\text{tr}A(x_\infty(H_c), y_\infty(H_c)) > 0$.

Proof. The proof depends on the fact that for H close to H_c there must be a (full) orbit connecting $(x_\infty(H), y_\infty(H))$ and $(\xi(H), \eta(H))$ as t runs from $-\infty$ to $+\infty$. This implies that there cannot be a closed orbit around $(x_\infty(H), y_\infty(H))$ and thus rules out the “good” unstable and “bad” asymptotically stable cases.

The details of the proof of the existence of a connecting orbit may be omitted without serious loss. To prove the existence of a connecting orbit, we proceed as follows. For $H < H_c$ but H sufficiently close to H_c , the continuity of $trA(x_0, y_0)$ in (x_0, y_0) and of $(x_\infty(H), y_\infty(H))$ in H implies that $trA(x_\infty(H), y_\infty(H)) \neq 0$. Again by continuity, there is a neighborhood of $(x_\infty(H), y_\infty(H))$ in which $trA(x, y) \neq 0$. We take a point C on $f(x, y) = 0$ above $(x_\infty(H), y_\infty(H))$ but close to $(x_\infty(H), y_\infty(H))$ and construct the orbit of $x' = xf(x, y), y' = yg(x, y) - H$ backward in time until it meets $yg(x, y) = H$ in a point B . We may assume that B is close to $(x_\infty(H), y_\infty(H))$, for if it is not we can take a point B' on $yg(x, y) = H$ close to $(x_\infty(H), y_\infty(H))$ and construct the orbit forward in time until it meets $f(x, y) = 0$ closer to $(x_\infty(H), y_\infty(H))$ than C , and use B' and C' in place of B and C . Now continue the orbit through B backward in time to a point A below and to the right of the saddle point $(\xi(H), \eta(H))$ and above the curve $yg(x, y) = H$. This is possible unless the orbit meets $f(x, y) = 0$ between $(x_\infty(H), y_\infty(H))$ and $(\xi(H), \eta(H))$, in which case we choose H larger so that $(\xi(H), \eta(H))$ moves up to the point on $f(x, y) = 0$ where the orbit meets it; increasing H reduces y' , and since $x' < 0$, this flattens the orbit and moves the intersection of the orbit with $f(x, y) = 0$ down below $(\xi(H), \eta(H))$. Thus we may assume that A is below and to the right of the saddle point. By an argument similar to that used to show that B may be assumed close to $(x_\infty(H), y_\infty(H))$, we may assume that A is close to $(\xi(H), \eta(H))$. Now draw a vertical line through C and a horizontal line through A , meeting in D . For H sufficiently close to H_c , the points $(x_\infty(H), y_\infty(H))$ and $(\xi(H), \eta(H))$ are close together, and we may assume that in the region $\mathcal{N}$ bounded by the arc ABC , the vertical line segment CD , and the horizontal line segment DA , we have $trA(x, y) \neq 0$. By the Bendixson criterion there can be no closed orbit contained in $\mathcal{N}$, and no homoclinic orbit from the saddle point $(\xi(H), \eta(H))$ contained in $\mathcal{N}$ (Figure 6.13).

By the Poincaré–Bendixson theorem, a nonconstant orbit that remains in $\mathcal{N}$ for $-\infty < t < \infty$ must have the equilibrium $(x_\infty(H), y_\infty(H))$ and the saddle point $(\xi(H), \eta(H))$ as its limiting sets for $t \rightarrow \pm\infty$ and must therefore be a connecting orbit. Thus if there is no connecting orbit, then every orbit starting in $\mathcal{N}$ must leave $\mathcal{N}$. We know that one of the unstable separatrices at the saddle point leaves $\mathcal{N}$ (downward). If the other one also leaves $\mathcal{N}$ as $t \rightarrow \infty$, it must do so across the line segment DA . But then there is an asymptotically stable separatrix at the saddle point between these two unstable separatrices that cannot leave $\mathcal{N}$ as $t \rightarrow -\infty$ because it could leave $\mathcal{N}$ only across the line segment DA and such a crossing would be in the wrong direction. Thus there must be a separatrix that remains in $\mathcal{N}$ for $-\infty < t < \infty$, and this is the desired connecting orbit. We have now established Theorem 6.1.

The value H_c is the maximum harvest rate for which coexistence is possible. However, if the system is in the “bad” case (unstable) at H_c , there must in fact be a value $H^* < H_c$ such that coexistence is possible only for $H \leq H^*$. Then H^* is the true maximum harvest rate. The value H_c may be calculated by looking for equilibria,

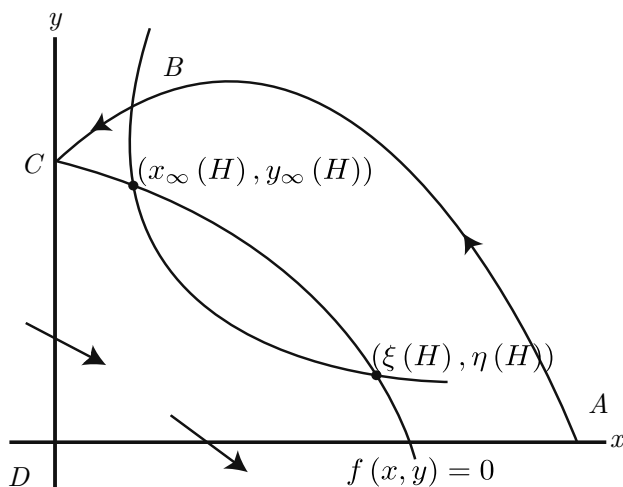


Fig. 6.13 No homoclinic orbit.

but to find H^* it is necessary to compute orbits near the saddle point and determine whether as $t \rightarrow \infty$ they are unbounded (“good”) or go to $(x_\infty(H), y_\infty(H))$ (“bad”).

Example 1: Consider the model of Rosenzweig–MacArthur type

$$x' = rx \left(1 - \frac{x}{K}\right) - \frac{cxy}{x+A}, \quad (6.2)$$

$$y' = sy \left(\frac{x}{x+A} - \frac{J}{J+A}\right) - H = \frac{sA}{J+A} \cdot \frac{x-J}{x+A} - H, \quad (6.3)$$

in which the prey satisfies a logistic equation in the absence of predators and $\frac{cx}{x+A}$ is the *predator functional response*—the rate at which predators remove prey. Here, A represents a prey population level at which the predators’ attack capability begins to saturate, K is the equilibrium population size for prey in the absence of predators, and J is the prey population size in equilibrium with predators without harvesting. For this model we may calculate that

$$x_\infty(H) = \frac{1}{2} \left((K+J) - \sqrt{(K-J)^2 - \frac{4K(J+A)}{rsA} H} \right),$$

$$y_\infty(H) = \frac{r}{K} (x_\infty(H) + A) (K - x_\infty(H)).$$

In particular, $x_\infty(0) = J$ and $x_\infty(H_c) = \frac{1}{2}(K+J)$, with $H_c = \frac{rsA(K-J)^2}{4K(J+A)}$. An equilibrium at $(x_\infty(H), y_\infty(H))$ is asymptotically stable if and only if

$$sAK(x_\infty(H) - J) < rx_\infty(H)(J+A)(2x_\infty(H) + A - K).$$

For example, if $A = 10$, $J = 20$, $K = 40$, $c = 1$, $r = 1$, $s = 1$, we have $H_c = 0.833$. Calculation shows that the equilibrium is asymptotically stable for $0 \leq H < H_c$, and numerical simulation of orbits shows that the system is in the “good” case for $0 \leq H < H_c$.

If $A = 10$, $J = 20$, $K = 40$, $c = 1$, $r = 1$, $s = 7$, we have $H_c = 5.833$. Calculation shows that the equilibrium is asymptotically stable for $0 \leq H < 5.185$, and simulation shows a transition from “good” to “bad” case at $H = 5.185$. Thus $H^* = 5.185$, the effective maximum harvest.

If $A = 10$, $J = 20$, $K = 45$, $c = 1$, $r = 1$, $s = 5$, we have $H_c = 5.787$. Calculation shows that the equilibrium is asymptotically stable if $0 \leq H < 3.704$ or $5.556 < H < H_c$. Thus if $3.704 < H < 5.556$ there is an asymptotically stable limit cycle; otherwise, there is an asymptotically stable equilibrium.

If $A = 10$, $J = 20$, $K = 60$, $c = 1$, $r = 2$, $s = 1$, we have $H_c = 4.444$. Calculation shows that the equilibrium is unstable for $0 \leq H < 2.322$, and asymptotically stable for $2.322 < H < H_c$. Thus there is an asymptotically stable limit cycle for small H that disappears when $H = 2.322$ and is replaced by an asymptotically stable equilibrium.

A final example is $A = 10$, $J = 20$, $K = 60$, $c = 1$, $r = 1$, $s = 4$, for which $H_c = 8.889$. Calculation shows that the equilibrium is unstable for all H and simulation indicates a transition from “good” to “bad” cases for $H = 0.785$. Thus there is an asymptotically stable limit cycle for $0 \leq H < 0.785$ and predator extinction for all harvest rates greater than $H^* = 0.785$. Observe that the true maximum harvest rate is much less than the critical harvest rate H_c , which can be calculated from equilibrium analysis.

Exercises

In each of Exercises 1–5, find the critical harvest rate of the system (6.2) with the given parameter values, find the range of harvest rates for which there is an asymptotically stable equilibrium, and carry out numerical simulations using a computer algebra system to describe the behavior of the system.

1. $A = 10$, $J = 20$, $K = 40$, $c = 1$, $r = 1$, $s = 1$.
2. $A = 10$, $J = 20$, $K = 40$, $c = 1$, $r = 1$, $s = 7$.
3. $A = 10$, $J = 20$, $K = 45$, $c = 1$, $r = 1$, $s = 5$.
4. $A = 10$, $J = 20$, $K = 60$, $c = 1$, $r = 2$, $s = 1$.
5. $A = 10$, $J = 20$, $K = 60$, $c = 1$, $r = 1$, $s = 4$.

6.3 Intermittent Harvesting of Predator–Prey Systems

The harvesting of predators in a predator–prey system at a rate u that may depend on time is modeled by the system

$$x' = xf(x, y), \quad y' = yg(x, y) - u(t).$$

Let us assume that the harvest rate is constrained by a condition $0 \leq u(t) \leq H$. Our goal is to determine a harvesting policy, or choice of u as a function of t , that will maximize the long-term average yield from harvesting,

$$Y = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t u(\tau) d\tau,$$

but we will also require that u be chosen in such a way as to avoid the collapse of the system by demanding that $x(t) > 0$, $y(t) > 0$ for $0 \leq t < \infty$.

The problem that we have formulated is an *optimal control* problem with an objective function $Y[u]$ that is linear in the control variable. It is an immediate consequence of the general theory of, specifically the Pontryagin maximum principle [Pontryagin, et al, (1962)] that the optimal control must be a “bang-bang” control, that is, a control that switches between the values $u = 0$ and $u = H$, possibly combined with some time intervals on which the control is “singular.”

If an optimal control makes only a finite number of switches between the values $u = 0$ and $u = H$, then we must have $u = H$ for all large t , and this implies that the yield Y is equal to H . As we have seen in the preceding section, there is a value $H_c = \max_{f(x,y)=0} yg(x, y)$ with the property that if $H > H_c$, then every solution of $x' = xf(x, y)$, $y' = yg(x, y) - H$ reaches the x -axis in finite time. Thus an optimal control can make a finite number of switches only if $H \leq H_c$. Even for $H < H_c$, the system may collapse if it is in the “bad” case, which occurs if its equilibrium is unstable as $H \rightarrow H_c$, and then there is a value $H^* < H_c$ such that the system actually collapses for $H > H^*$. If the equilibrium is asymptotically stable as $H \rightarrow H_c$ then for every $H < H_c$ it is possible to bring the solution of $x' = xf(x, y)$, $y' = yg(x, y) - u(t)$ into the set of initial conditions for which the solution of $x' = xf(x, y)$, $y' = yg(x, y) - H$ tends to the equilibrium by making a finite number of switches between $u = 0$ and $u = H$. Thus for such a system we can obtain a yield H for every $H \leq H_c$ with a finite number of switches.

In considering controls with an infinite number of switches between $u = 0$ and $u = H$, it would be desirable for practical applications to have a reasonably simple criterion for determining switching points. One very simple control strategy would be to assign upper and lower thresholds y_U and y_L for predator population size, start the system at $t = 0$ with $y = y_L$ and $u = 0$, and switch from $u = 0$ to $u = H$ whenever y increases through y_U and switch from $u = H$ to $u = 0$ whenever y decreases through y_L . This procedure is safe, in the sense that it prevents predator extinction. Numerical simulations indicate that it is always possible to choose the parameters H , y_U , and y_L so as to make the yields Y arbitrarily close to H_c , even for systems that are unstable at H_c and therefore collapse for smaller yields under constant yield harvesting. For such systems there is a tendency for the orbit to approach a periodic orbit with two switch points per cycle (Figure 6.14).

There may be an interval of values of H for given y_U and y_L on which the behavior is more complicated, with period-doubling indicated by periodic orbits having four or eight switch points per cycle, and apparent chaotic behavior (Figure 6.15).

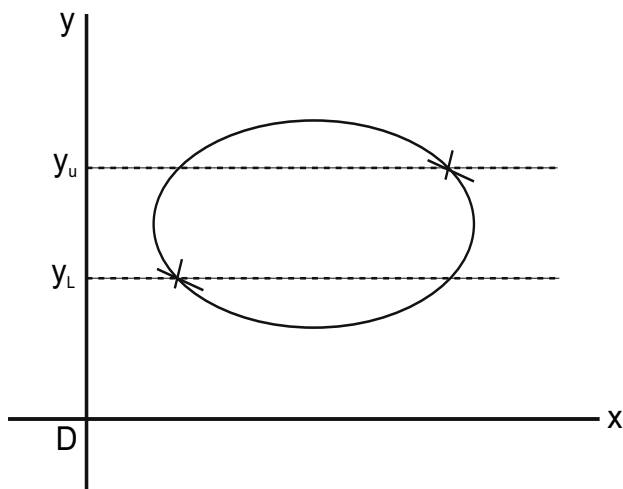


Fig. 6.14 Periodic orbit with switching.

Such behavior appears to occur only for $H < H_c$ and with yields that are not close to optimal.

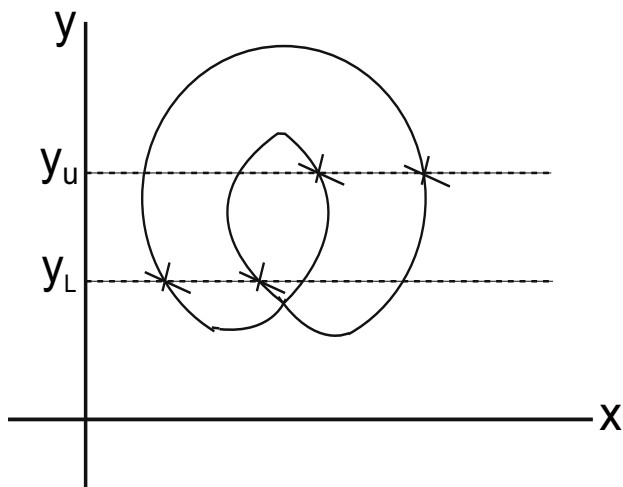


Fig. 6.15 Multiple switch points.

It is possible to prove that if the functions $f(x,y)$ and $g(x,y)$ satisfy some additional concavity conditions (always satisfied by Rosenzweig–MacArthur models and by Kolmogorov models with $g(x,y)$ independent of y and $f(x,y)$ linear in y), then if the control that maximizes the yield gives either a periodic solution of the controlled system or a solution that tends to an equilibrium, then the yield cannot be

greater than H_c for any H . If $H < H_c$, it is clear that the yield cannot be greater than H_c . For $H > H_c$ it appears plausible that optimal controls lead to periodic orbits and therefore that the yield cannot be greater than H_c .

If no bound $u(t) \leq H$ is imposed on the harvest rate, then the analogue of a bang-bang control that switches between the value $u = 0$ and the maximum permissible value of u would be an *instantaneous harvest*. We may model this procedure by setting an upper threshold y_U and a lower threshold y_L and considering only the unharvested system $x' = xf(x, y)$, $y' = yg(x, y)$; whenever $y(t)$ reaches y_U we remove $(y_U - y_L)$ predators to make $y(t) = y_L$ and restart the system with the same x -value.

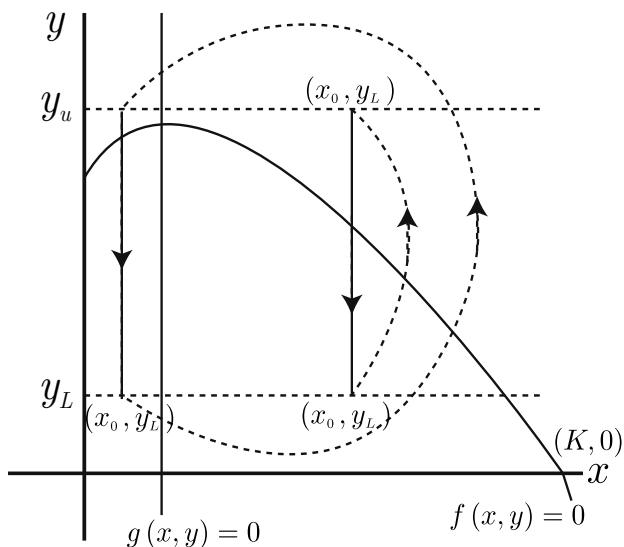
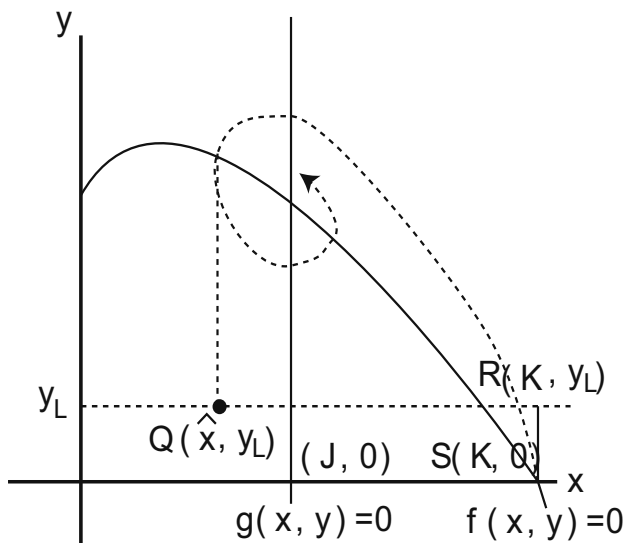
Another instantaneous harvesting policy that occurs more commonly in practice is to set a fixed “closed season” of length T at the end of which the predator population is harvested instantaneously down to a set escapement level y_L . These two policies are equivalent mathematically; the only difference is that in the first, y_U and y_L are assigned and T is to be determined, while in the second, y_L and T are assigned and y_U is determined from the orbit. Another equivalent variant would assign T and $(y_U - y_L)$ with y_L or y_U to be determined from the orbit. In practice, there may be a very short hunting season and a relatively long closed season, and instantaneous harvesting approximates this procedure.

For instantaneous harvesting, the following convergence result holds.

Theorem 6.2. *Let the value y_L be small enough that the line $y = y_L$ and the curve $f(x, y) = 0$ intersect. If $y_U \geq y_L$ is sufficiently close to y_L , then the instantaneous harvesting procedure converges to a periodic orbit consisting of a solution of $x' = xf(x, y)$, $y' = yg(x, y)$ from a point (x_0, y_L) to a point (x_0, y_U) and a vertical line segment from (x_0, y_U) to (x_0, y_L) (see Figure 6.16).*

Proof. The unharvested system has a unique equilibrium P_∞ in the interior of the first quadrant which may be asymptotically stable or unstable, and a saddle point $S(K, 0)$. There is a full orbit from S to P_∞ or to a limit cycle around P_∞ for $-\infty < t < \infty$. Let $\hat{x}$ be the minimum value of x on this orbit and let Q be the point $(\hat{x}, y_L)$, R the point (K, y_L) (Figure 6.17).

For any solution $(x(t), y(t))$, if $x(0) < x_L, y(0) = y_L$, then $x(t) > x(0)$ for $0 < t < \infty$, and if $x(0) > K, y(0) = y_L$, then $x(t) < x(0)$ for $0 < t < \infty$. The verification of these facts depends on the observation that no other orbit can cross the orbit from S to P_∞ or a limit cycle around P_∞ . It now follows that for each $T > 0$ there is an x_0 with $\hat{x} < x_0 < K$ such that if $x(0) = x_0, y(0) = y_L$, then $x(T) = x(0) = x_0$. Further, the policy of starting the system at a point (x^*, y_L) at $t = 0$, letting it run to $(x(T), y(T))$, and then harvesting $y(T) - y_L$ predators to bring the system to the point $(x(T), y_L)$ will bring $x(T)$ closer to x_0 than x^* is. Thus repetition of the process must give convergence to a periodic orbit starting at (x_0, y_L) , following the orbit of the system to $(x_0, y(T))$ at time T , and then going vertically to (x_0, y_L) . The procedure of setting $y_U = y(T)$ rather than specifying T gives the same periodic orbit, which then determines T , provided y_U is close enough to y_L that the orbit starting at (x_0, y_L) reaches the horizontal line $y = y_U$. This completes the proof of the theorem.

**Fig. 6.16** Instantaneous harvesting.**Fig. 6.17** Instantaneous harvesting.

The yield Y under instantaneous harvesting is given by $Y = \frac{y_U - y_L}{T}$. Since the solution of the controlled system is periodic, it follows from the results cited earlier in this section that this yield is at most H_c . Obviously, it is possible to make the yield arbitrarily close to H_c by making T very small and arranging that the orbit lie very close to $(x_\infty(H_c), y_\infty(H_c))$ so that $y(t)g(x(t), y(t))$ is very close to H_c for all

points of the orbit. Since the yield is the average of $y'(t) = y(t)g(x(t), y(t))$ over $0 \leq t \leq T$, this yield is close to H_c . Of more interest is the fact that numerical simulations indicate that yields arbitrarily close to H_c can always be obtained for a range of values of T not confined to very small values. This is true in the “bad” case, in which such yields cannot be achieved by constant-yield harvesting.

The material in this section is from [Brauer & Soudack (1979)], where additional discussion may be found.

Exercises

1. Consider the system

$$\begin{aligned}x' &= x\left(1 - \frac{x}{60}\right) - \frac{xy}{x+10}, \\y' &= 4y\left(\frac{x}{x+10} - \frac{2}{3}\right) = \frac{4y(x-20)}{3(x+10)}.\end{aligned}$$

We will choose $y_L = 5$, $T = 4$. Use a computer algebra system to simulate this system with $x(0) = 40$, $y(0) = 5$ for $0 \leq t \leq 4$. After recording the instantaneous harvest $y(4) - 5$, restart with initial values $(x(4), 5)$ at $t = 4$ and run for $4 \leq t \leq 8$. Repeat this process several times and estimate the average instantaneous harvest.

6.4 Some Economic Aspects of Harvesting

The incorporation of economic considerations into resource-harvesting models leads to a rather new subject, bioeconomics. We shall only scratch the surface of this subject, considering some simple fisheries models and introducing the idea of an optimal control problem and illustrating the use of the maximum principle of Pontryagin (1962). For further exploration of bioeconomics, we refer the reader to the standard text reference by Clark (1990).

Let us consider a model for an open-access fishery, that is, a fishery in which the exploitation of resources is completely uncontrolled. Our model will be an idealization, but in practice, fisheries are usually relatively unregulated. However, the abuses of open access exploitation have led to the imposition of regulations in some instances.

We consider a fish population that is governed by a first-order differential equation,

$$y' = F(y),$$

and which is subjected to constant-effort harvesting, so that the model becomes

$$y' = F(y) - Ey. \quad (6.4)$$

The *yield*, or harvest in unit time, is

$$Y(E) = Ey.$$

Corresponding to an effort E there is an equilibrium population size $y_\infty = y_\infty(E)$ given by

$$F(y_\infty) = Ey_\infty$$

for $0 \leq E \leq E^* = F'(0)$. As we have seen in Section 1.5, we may plot $Y(E)$ as a function of E to obtain the *yield-effort curve* (Figure 6.18).

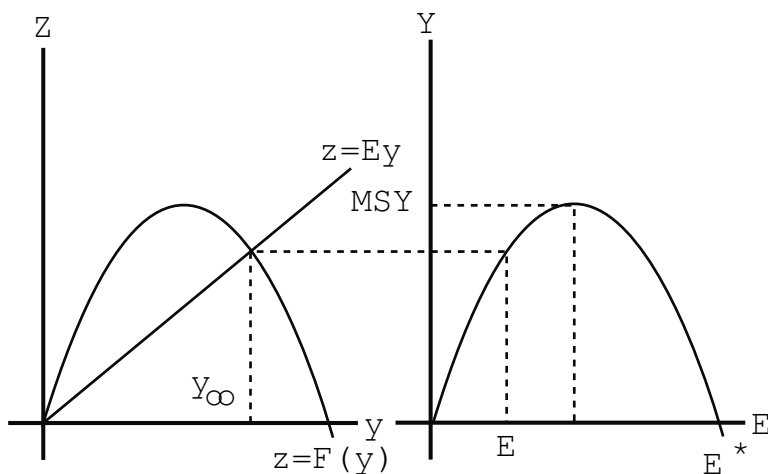


Fig. 6.18 The yield-effort curve.

In Figure 6.18, the yield-effort curve is shown for a compensation model, rising continuously to a maximum, called the *maximum sustainable yield* (MSY) and then decreasing continuously to zero for $E = E^*$. Recall from Section 1.5, however, that for a depensation model there may be a critical effort E^* at which the yield drops to zero discontinuously.

Example 1: For logistic growth, $F(y) = ry \left(1 - \frac{y}{K}\right)$ we have

$$\begin{aligned} F(y_\infty(E)) &= ry_\infty(E) \left[1 - \frac{y_\infty(E)}{K}\right] = Ey_\infty(E), \\ r \left[1 - \frac{y_\infty(E)}{K}\right] &= E, \\ y_\infty(E) &= K \left(1 - \frac{E}{r}\right). \end{aligned}$$

Thus

$$Y(E) = Ey_{\infty}(E) = KE \left(1 - \frac{E}{r}\right).$$

The maximum sustainable yield (MSY) is obtained by solving $\frac{dY}{dE} = 0$, giving $E_{MSY} = \frac{r}{2}$, $y(E_{MSY}) = \frac{K}{2}$, and $Y(E_{MSY}) = \frac{rK}{4}$.

Example 2: For the Ricker model $F(y) = rye^{-ay}$, we have

$$re^{-ay_{\infty}(E)} = E,$$

which leads to

$$y_{\infty}(E) = \frac{1}{a} \log \frac{r}{E} \quad \text{and} \quad Y(E) = \frac{E}{a} \log \frac{r}{E}.$$

Then

$$Y'(E) = \frac{E}{a} \log \frac{r}{E} - \frac{1}{a},$$

and maximum sustainable yield is obtained when $E \log \frac{r}{E} = 1$, which gives $Y(E_{MSY}) = \frac{1}{a}$.

To incorporate the economic aspects of a harvesting model we must attach a revenue from the sale of this fish or other item being harvested as well as a cost to carry out the harvesting. The simplest assumptions would be that the *cost* C of fishing is proportional to the effort,

$$C = cE,$$

and that the price does not depend on the supply of fish available to sell, so that the *revenue* R is a constant multiple of the quantity harvested,

$$R = pY(E).$$

This would give a *sustainable economic rent*, meaning a profit per unit time that can be maintained indefinitely, of

$$R - C = rY(E) - cE.$$

More generally, economic models may assume a functional relation, known as a *demand curve*, between the unit selling price p and the quantity Y available for sale. It would be reasonable to assume that p is a nonincreasing function of Y , or equivalently, that Y is a nonincreasing function of p .

A basic (but considerably oversimplified) principle is that in an open access fishery the effort tends to approach an equilibrium effort E_{∞} , called the *bionomic equilibrium effort*, at which the sustainable economic rent is zero. The argument is that at the bionomic equilibrium, $R = C$, and that $R > C$ if $E < E_{\infty}$, $R < C$ if $E > E_{\infty}$. If $E > E_{\infty}$, then $R < C$, and sustainable economic rent is negative. This means that some fisheries are losing money and therefore drop out of the market, thus decreas-

ing the total effort. If $E < E_\infty$, then $R > C$, and fishing is profitable, encouraging new fishers to join the market and increase the total effort. Thus there should be a tendency to approach the bionomic equilibrium, although we should recognize that the delay that represents the time needed to respond to the state of the market could produce oscillations around the equilibrium (Figure 6.19).

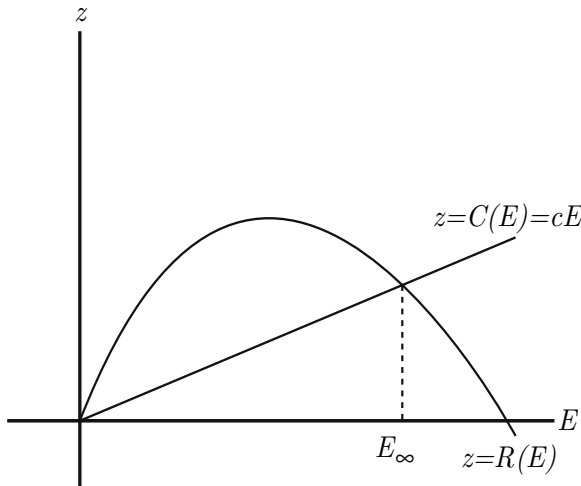


Fig. 6.19 Bionomic equilibrium.

There are many oversimplifications in this bionomic equilibrium principle. For example, the cost of fishing should also include the loss from not undertaking some alternative activity (opportunity cost), which in practice might be the most important component of fishing cost. If opportunity cost is included in the total cost, then the fishers who remain in an open access fishery would tend to be those with the poorest alternate economic opportunities.

Bionomic equilibrium is a situation in which there is *economic overfishing*, because a fishery that would produce positive economic rent if fishing effort were decreased is producing zero economic rent. Another possibility is *biological overfishing*, where $E_\infty > E_{MSY}$ and thus the sustainable yield and equilibrium population size are smaller than they would be if the effort were decreased.

Example 1: Consider the logistic model

$$y' = ry \left(1 - \frac{y}{K} \right) - Ey,$$

with $R = py_\infty(E)$ and $C = cE$. At equilibrium, $y_\infty = \frac{K}{r}(r - E)$, $R = \frac{pKE}{r}(r - E)$, and $C = cE$. Then the bionomic equilibrium E_∞ is determined from

$$\frac{pKE}{r}(r - E) = cE,$$

which gives

$$pK(r - E) = rc \quad \text{and} \quad r - E = \frac{rc}{pK},$$

so that

$$E_\infty = r - \frac{rc}{pK} = r \left(1 - \frac{c}{pK} \right).$$

The corresponding equilibrium population size is

$$y_\infty = K \left(1 - \frac{E_\infty}{r} \right) = K \left[1 - \left(1 - \frac{c}{pK} \right) \right] = \frac{c}{p}.$$

Thus if $\frac{c}{p} > K$, there is no exploitation at the bionomic equilibrium. If $\frac{c}{p} < \frac{K}{2}$, there is biological overfishing at bionomic equilibrium, but so long as $p > 0$, we do not have biological extinction at bionomic equilibrium. The bionomic equilibrium depends upon the cost–price ratio $\frac{c}{p}$ as well on the biological parameters r, K . In fact, under the assumptions of constant unit price p and constant cost c of unit effort, the equilibrium population size at bionomic equilibrium is equal to the cost–price ratio, and the harvesting effort at bionomic equilibrium may be found by simply calculating the effort that leads to this equilibrium population size.

Exercises

1. The model for fisheries

$$\frac{dx}{dt} = a - bx$$

was proposed by Schoener in 1973. Determine the equation of the yield–effort curve and explain its peculiar feature.

2. Find x_{MSY} (density of maximum sustainable yield) and $F(x_{MSY})$ (the maximum sustainable yield) for the Gompertz law of population growth:

$$\frac{dx}{dt} = rx \log \left(\frac{K}{x} \right).$$

Also sketch the yield–effort curve. What is E^* (effort at MSY)?

3. Find $x_{\max}$ (density of maximum sustainable yield) and $F(x_{\max})$ (the maximum sustainable yield) for a population governed by the equation

$$\frac{dx}{dt} = rxe^{-x} - dx$$

with $r > d$.

4. What happens to the yield in a population governed by the equation

$$\frac{dx}{dt} = rxe^{-x} - dx$$

with $r < d$?

5. Show that if a population is governed by the equation

$$\frac{dx}{dt} = r\sqrt{x},$$

there is no maximum sustainable yield.

6.5 Optimization of Harvesting Returns

Consider an open-access fishery that is in bionomic equilibrium with biological overfishing. Let us suppose that we try to regulate this fishery by reducing the harvesting effort in order to increase the yield. The immediate effect of reducing effort is to reduce the yield, but the long-term effect will be to increase the yield by increasing the equilibrium population level. We would like to determine the optimal trade-off between current and future harvesting revenue. In order to do this we must take into account the time value of money because of interest. Since the quantity that we shall study is the present value of all future harvest rents, we shall do this by using a discount rate δ , and since we are studying a continuous model, we shall assume that this discount rate is compounded continuously.

When we speak of a discount rate δ , we mean that the present value of an amount that would have value 1 at a time t units in the future is $(1 - \delta)^t$. Normally the time unit is one year. If the discount is *compounded* K times per year, we use a discount rate of $\frac{\delta}{K}$ per discount period. Then t years would be tK discount periods, and the present value of an amount 1 at a time t units in the future would be $(1 - \frac{\delta}{K})^{tK}$. By continuous compounding, we mean that the present value would be the limit of this quantity as $K \rightarrow \infty$. Because

$$\lim_{K \rightarrow \infty} \left(1 - \frac{\delta}{K}\right)^{Kt} = e^{-\delta t},$$

the effect of a continuously compounded discount rate δ would be to introduce a factor $e^{-\delta t}$ for a rent at time t years in the future.

We now generalize the model (6.4) slightly by assuming that the harvest at time t is proportional to the effort E as before, but is not necessarily proportional to the population size $y(t)$. Thus we write the harvest at time t as

$$h(t) = EG\{y(t)\}, \quad (6.5)$$

where $G(y)$ is a nonnegative, nondecreasing function of y . We continue to assume a constant unit price p and a constant cost c of unit effort. Then the unit harvest cost when the population level is y is a function $c(y)$ given by

$$c(y) = \frac{c}{G(y)}. \quad (6.6)$$

An input of effort $E\Delta t$ from time t to time $t + \Delta t$ produces a harvest $EG\{y(t)\}\Delta t$, which produces a revenue of $pEG\{y(t)\}\Delta t$ at a cost of $cE\Delta t$. Thus the net revenue is

$$pEG\{y(t)\}\Delta t - cE\Delta t = [p - c\{y(t)\}]h(t)\Delta t, \quad (6.7)$$

making use of (6.5) and (6.6). The present value of all future harvesting effort is thus

$$P = \int_0^{\infty} e^{-\delta t} [p - c\{y(t)\}]h(t)dt. \quad (6.8)$$

Note that we must use a positive discount rate δ , since otherwise the infinite integral in (6.8) would diverge. If we choose to use a *finite time horizon* by considering only the harvesting effort until time T , then we would replace (6.8) by

$$P = \int_0^T e^{-\delta t} [p - c\{y(t)\}]h(t)dt. \quad (6.9)$$

With a finite time horizon we could choose to disregard the time value of money by taking $\delta = 0$ without introducing any problems with divergence of the integral.

The optimal control problem that we wish to solve is to choose the harvest rate function that will maximize the integral (6.8) subject to the *state equation*

$$y' = F(y) - h(t) = F\{y(t)\} - EG\{y(t)\} \quad (6.10)$$

and the constraints $y(t) \geq 0$ (because a realistic model requires a nonnegative population level) and $0 \leq h(t) \leq h_{\max}$ (because a fishery always has a maximum harvesting capability).

The solution of this optimal control problem, which we shall justify later, is that we should first choose the equilibrium population level y^* as the solution of the equation

$$F'(y) - \frac{c'(y)F(y)}{p - c(y)} = \delta,$$

or equivalently,

$$\frac{d}{dy}[F(y)\{p - c(y)\}] = \delta[p - c(y)]. \quad (6.11)$$

We then choose the effort E so that

$$h = EG(y^*) = F(y^*).$$

This still leaves open the question of the optimal way to reach this *optimal equilibrium population* level y^* . The answer to this question is that we should use the harvest rate $h^*(t)$ at time t that drives $y(t)$ to y^* as rapidly as possible,

$$h^*(t) = \begin{cases} h_{\max} & (y > y^*), \\ F(y^*) & (y = y^*), \\ 0 & (y < y^*). \end{cases}$$

This type of control, taking the extreme values 0 and $h_{\max}$ and switching when $y = y^*$, is called a *bang-bang control*. It may appear to be an extreme policy in practice, since it involves closing down a fishery completely, possibly for a long period. Such a policy is a consequence of the assumed linear dependence on E . In the next section we shall consider a nonlinear control problem for which the optimal control is less extreme.

At equilibrium of the fishery model (6.10), $F(y) = h$. Thus the sustainable economic rent at y is equal to

$$\rho(y) = [p - c(y)]F(y).$$

With this definition of $\rho(y)$, we may write the optimizing condition (6.11) as

$$\rho'(y) = \delta[p - c(y)]. \quad (6.12)$$

A marginal decrease of Δy in y gives an immediate rent of $[p - c(y)]\Delta y$ and decrease in sustained rent

$$\Delta p \approx \rho'(y)\Delta y$$

with a present value of

$$\int_0^\infty e^{-\delta t} \rho'(y)\Delta y dt = \frac{1}{\delta} \rho'(y)\Delta y.$$

Thus for the optimal solution, the marginal immediate gain is equal to the present value of the marginal future loss.

From (6.6) and the assumption that $G(y)$ is a nondecreasing function of y , it follows that $c(y)$ is nonincreasing, and $p - c(y)$ is nondecreasing. If $\rho'(y)$ is a decreasing function, the equation (6.12) has a unique solution (Figure 6.20).

The significance of $\delta = 0$ is that we should maximize the sustained economic rent, that is, we should maximize ρ by setting $\rho'(y) = 0$. The corresponding population level is y_0 . If we let $\delta \rightarrow \infty$, we set zero value on future revenues and set $p - c(y) = 0$ (no sustainable rent). The corresponding population level is y_∞ . Here y_0 is defined by $\rho'(y) = 0$ and y_∞ by $c(y_\infty) = p$; we have $y_\infty < y^* < y_0$.

The process described above is for a harvest $h(t) = EG\{y(t)\}$, which is not necessarily proportional to population size. In the special case $h(t) = Ey(t)$ that we considered earlier, (6.6) implies $c(y) = \frac{c}{y}$, and this is used in the determination of y^* . Once y^* has been determined, it is easy to calculate $F(y^*)$ and thus to determine the switching point in $h^*(t)$.

Example 1: Consider a logistic model with constant-effort harvest, so that

$$F(y) = ry \left(1 - \frac{y}{K}\right), \quad G(Y) = y, \quad c(y) = \frac{c}{y}.$$

The condition (6.12) becomes

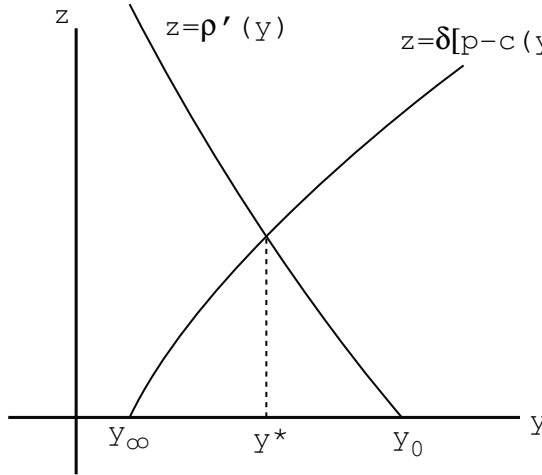


Fig. 6.20 Unique optimal equilibrium.

$$\begin{aligned}
 r \frac{d}{dy} \left[y \left(p - \frac{c}{y} \right) \left(1 - \frac{y}{K} \right) \right] &= \delta \left(p - \frac{c}{y} \right), \\
 r \frac{d}{dy} \left[y \left(p - \frac{c}{y} \right) - \frac{p}{K} y^2 + \frac{c}{K} y \right] &= \delta \left(p - \frac{c}{y} \right), \\
 2rpy^2 + y(\delta pK - rpK - rc) - \delta cK &= 0.
 \end{aligned}$$

This is a quadratic equation having one positive root and one negative root, and we take y^* to be the positive root,

$$y^* = \frac{K}{4} \left[\left(\frac{c}{pK} + 1 - \frac{\delta}{r} \right) + \left\{ \left(\frac{c}{pK} + 1 - \frac{\delta}{r} \right)^2 + \frac{8c\delta}{prK} \right\}^{\frac{1}{2}} \right].$$

For zero discount, $\delta = 0$, we obtain $y_0 = \frac{K}{2} \left(\frac{c}{pK} + 1 \right)$, and since $y_\infty = \frac{c}{p}$, we have

$$y_0 = \frac{1}{2}(y_\infty + K) = \frac{1}{2}y_\infty + y_{MSY} > y_{MSY}.$$

Thus we do not have biological overfishing if $\delta = 0$. However, increasing δ will decrease y^* from y_0 . An extreme case is $y_\infty = 0$, when fishing has zero cost, and then

$$y^* = \frac{K}{2} \left(1 - \frac{\delta}{r} \right) < y_{MSY}.$$

If the bionomic growth ratio $\frac{\delta}{r}$ exceeds 1, then $y^* = 0$, and the optimal harvest policy leads to the most rapid possible extinction of the resource. The interpretation is that if the discount rate δ is greater than the intrinsic growth rate r , the fisher should cash

in the resources as quickly as possible. This optimal fishing policy may of course not be considered ideal by those with concerns other than the economic rent of the fishery.

Exercise

1. For a population governed by the Gompertz law of population growth

$$\frac{dx}{dt} = rx \log\left(\frac{K}{x}\right)$$

find the bioeconomic equilibrium for a constant price p and a cost cE with c constant. Show that there is biological overfishing if $\frac{c}{p} < \frac{K}{e}$, but no biological extinction at bioeconomic equilibrium.

6.6 Justification of the Optimization Result

In the preceding section we stated a result that we used to provide the solution of an optimization problem. In this section we shall justify this result. In fact, we consider the more general

$$J[u] = \int_{t_0}^{t'} [f_0(t, y(t)) + g_0(t, y(t))u(t)] dt, \quad (6.13)$$

subject to a state equation

$$y' = f_1(t, y) + g_1(t, y)u, \quad (6.14)$$

with given values $y(t_0) = y_0$, $y(t_1) = y_1$, and with the *control* $u(t)$ chosen from an *admissible set* of functions that are piecewise continuous and satisfy

$$u_m \leq u(t) \leq u_M.$$

The problem considered in Section 6.6 was the special case

$$\begin{aligned} f_0(t, y) &= 0, & g_0(t, y) &= e^{-\delta t} [p - c(y)], \\ f_1(t, y) &= F(y), & g_1(t, y) &= -1, \\ u(t) &= h(t). \end{aligned}$$

However, in this problem we did not impose a *terminal condition* $y(t_1) = y_1$; the terminal condition is essentially replaced by the requirement that $y(t)$ approach an equilibrium as $t \rightarrow \infty$.

We begin the solution by solving the state equation (6.14) for u and substituting the result into (6.13), obtaining a form

$$\int_{t_0}^{t'} [G(t, y) + H(t, y)y'] dt \quad (6.15)$$

to be maximized under constraints

$$A(y, t) \leq y'(t) \leq B(y, t). \quad (6.16)$$

Theorem 6.3. *Let $y^*(t)$ be the unique solution of*

$$\frac{\partial G}{\partial y} = \frac{\partial H}{\partial t} \quad (6.17)$$

and suppose

$$\begin{aligned} \frac{\partial G}{\partial y} &\leq \frac{\partial H}{\partial t} & \text{if } y < y^*(t) \\ \frac{\partial G}{\partial y} &\geq \frac{\partial H}{\partial t} & \text{if } y > y^*(t). \end{aligned}$$

Then the optimal solution is the “closest possible trajectory to $y^(t)$ ”, defined as follows:*

- (i) *If $y_0 > y^*(t_0)$, $y(t)$ uses the fastest possible descent $y' = A(y, t)$, $y(t_0) = y_0$ until $y^*(t)$ is reached.*
- (ii) *If $y_0 < y^*(t_0)$, $y(t)$ uses the fastest possible ascent $y' = B(y, t)$, $y(t_0) = y_0$ until $y^*(t)$ is reached.*

Similarly, if $y_1 \neq y^*(t_1)$, use the same scheme backward from t_1 , until $y^*(t)$ is reached. Between these two points where $y^*(t)$ is reached, say t_a when $y(t)$ starting at t_0 reaches $y^*(t)$ and t_b when $y(t)$ ending at t_1 reaches $y^*(t)$, take $y(t) = y^*(t)$ [Figure 6.21].

Proof. We write the objective functional as a line integral

$$\int_{t_0}^{t_1} [G(t, y) + H(t, y)y'] dt = \int_C (Gdt + Hdy),$$

where C is the curve $y = y(t)$, $t_0 \leq t \leq t_1$. Suppose $y(t)$ is the claimed optimal trajectory and $y_1(t)$ is an arbitrary admissible curve from (t_0, y_0) to (t_1, y_1) . Because of the constraint (6.16) satisfied by $y_1(t)$, we must have $y_1(t) \leq y(t)$ at least for $t_0 \leq t \leq t_a$. Suppose $y_1(t) < y(t)$ on $t_0 < t < t_R$, $y_1(t_R) = y(t_R)$. Then

$$\int_{t_0}^{t_R} [G(t, y) + H(t, y)y'] dt - \int_{t_0}^{t_R} [G(t, y) + H(t, y)y'] dt = \int_{\Gamma} (Gdt + Hdy), \quad (6.18)$$

where Γ is the closed curve PRS as shown in Figure 6.22.

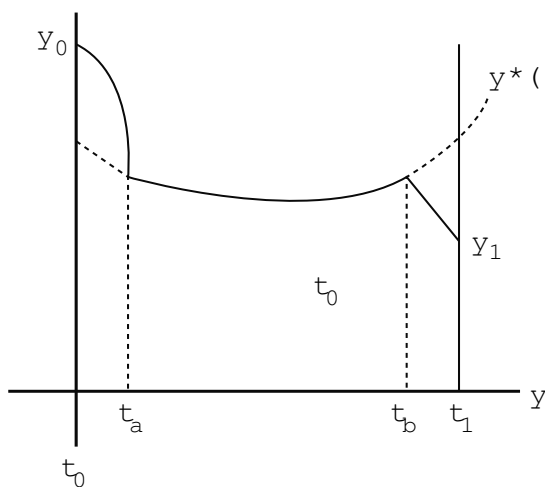


Fig. 6.21

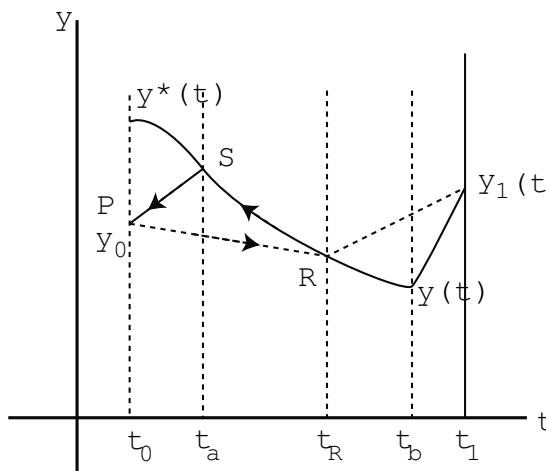


Fig. 6.22

By Green's theorem in the plane, this is equal to

$$\int_A \int \left(\frac{\partial G}{\partial y} - \frac{\partial H}{\partial t} \right) dy dt,$$

where A is the region in the interior of the curve Γ . Since this region lies below $y^*(t)$, $\frac{\partial G}{\partial y} \geq \frac{\partial H}{\partial t}$ in A , the integral (6.18) is nonnegative. This shows that $y(t)$ does better than $y_1(t)$ in optimizing on the interval $t_0 \leq t \leq t_R$, and a similar argument shows the optimality from t_R to t_1 . This completes the proof of the theorem. In fact, we

have established the stronger result that an optimal path must be optimal on every subinterval.

In the problem considered in Section 6.5, we used

$$u = h = F(y) - y',$$

$$J[h] = \int_0^\infty e^{-\delta t} [p - c\{y(t)\}][F\{y(t)\} - y'(t)] dt,$$

so that

$$G(t, y) = e^{-\delta t} [p - c(y)] F(y),$$

$$H(t, y) = -e^{-\delta t} [p - c(y)].$$

Then equation (6.17) becomes

$$e^{-\delta t} \frac{d}{dt} [\{p - c(y)\} F(y)] = \delta e^{-\delta t} [p - c(y)],$$

which is the same as equation (6.4) of Section 6.5, which we solved to determine the final value y^* .

If in the problem of Section 6.5 we had used a finite time horizon T instead of trying to optimize over $0 \leq t < \infty$ and had specified a terminal condition for $y(T)$, the theorem of this section would imply that it would be necessary to leave the path $y^*(t)$ before T and use a bang-bang control. Obviously, $h = 0$ would not be optimal, because $h = h_{max}$ for t near T would provide a positive contribution to the present value. The “horizon effect” of rapid resource depletion for t near T is an artificial consequence of the finite time horizon. The practical effect of this can be eliminated by taking a sufficiently remote horizon (letting $T \rightarrow \infty$). In nonlinear models, such as will be considered in the next section, horizon effects typically tend to zero exponentially for finite t as $T \rightarrow \infty$ and are effectively eliminated by taking T large.

6.7 A Nonlinear Optimization Problem

In this section we consider a model

$$y' = F(y) - h(t), \quad y(0) = y_0, \quad y(T) = y_T, \quad (6.19)$$

on a finite interval for which the revenue is not a linear function of h . In most real market situations the price p is not a constant but depends on the supply h ; it is reasonable to assume that $p(h)$ is a nonincreasing function of h . The revenue $R(h) = hp(h)$ is then not proportional to h . We will assume that the function $R(h)$ has the properties

$$R(h) > 0, \quad R'(h) \geq 0, \quad R''(h) \leq 0.$$

We will neglect the cost of harvesting, but an approach similar to the one we employ could treat a more general situation that does include harvesting cost. The objective function now is

$$J[h] = \int_0^T e^{-\delta t} R(h) dt. \quad (6.20)$$

The nonlinearity in h implies that the direct line integral approach used in the preceding section cannot be used. In its stead, we use the maximum principle of Pontryagin (1962).

The Pontryagin maximum principle deals with the general control problem of maximizing an objective functional

$$J[u] = \int_0^T g[t, y(t), u(t)] dt, \quad (6.21)$$

where $u(t)$ is chosen from some class of admissible controls u on $0 \leq t \leq T$, with a state equation

$$y'(t) = f[t, y(t), u(t)], \quad y(0) = y_0. \quad (6.22)$$

In addition, the control u must also be such that the response $y(t)$ satisfies the terminal condition

$$y(T) = y_T. \quad (6.23)$$

We define the *Hamiltonian*

$$\mathcal{H}[t, y(t), u(t), \lambda(t)] = g[t, y(t), u(t)] + \lambda(t) f[t, y(t), u(t)], \quad (6.24)$$

where λ is an unknown function called the *adjoint variable*. The maximum principle says that if u is the optimal control and y the corresponding response, then there is an adjoint variable λ such that

$$\frac{d\lambda}{dt} = -\frac{\partial \mathcal{H}}{\partial y} = -\frac{\partial g}{\partial y} - \lambda \frac{\partial f}{\partial y} \quad (6.25)$$

and the optimal control maximizes the value of the Hamiltonian over all admissible controls for each t , that is,

$$\mathcal{H}[t, y(t), u(t), \lambda(t)] = \max_{u \in U} \mathcal{H}[t, y(t), u(t), \lambda(t)]. \quad (6.26)$$

If the optimal control is in the interior of the control interval, that is, if the control constraints are not *binding*, then this maximization statement is

$$\frac{\partial \mathcal{H}}{\partial u} = 0. \quad (6.27)$$

In general, we have three functions $y(t)$, $u(t)$, and $\lambda(t)$ to determine, and three equations, namely the state equation (6.22), the adjoint equation (6.25), and the maximum statement (6.26) or (6.27) can be used for their determination.

For the linear problem solved in Section 6.5 of maximizing

$$\int_0^\infty e^{-\delta t} [p - c(y)] h(t) dt$$

with the state equation

$$y' = F(y) - h(t),$$

we have

$$g(t, y, h) = e^{-\delta t} [p - c(y)] h, \quad f(t, y, h) = F(y) - h(t),$$

in (6.21) and (6.22). Then (6.24) gives the Hamiltonian

$$\mathcal{H}[t, y, h, \lambda] = e^{-\delta t} [p - c(y)] h(t) + \lambda(t) [F(y) - h(t)],$$

and the adjoint equation (6.25) is

$$\frac{d\lambda}{dt} = -c'(y)h(t)e^{-\delta t} + \lambda F'(y). \quad (6.28)$$

If the constraints are not binding, that is, if we can maximize by taking $\frac{\partial \mathcal{H}}{\partial u} = 0$ without being concerned that the resulting u will not be an admissible control, the maximum condition (6.27) gives

$$e^{-\delta t} [p - c(y)] - \delta(t) = 0,$$

or

$$\lambda(t) = e^{-\delta t} [p - c(y)]. \quad (6.29)$$

Substituting the expression (6.29) into the adjoint equation (6.28), we have

$$\begin{aligned} -\delta e^{-\delta t} [p - c(y)] &= c'(y)h e^{-\delta t} - \lambda F'(y) \\ &= c'(y)F(y)e^{-\delta t} - e^{-\delta t} [p - c(y)]F'(y), \end{aligned} \quad (6.30)$$

or

$$F'(y) - \frac{c'(y)F(y)}{p - c(y)} = \delta,$$

which is the solution obtained in Section 6.5. Note that we in effect set an equilibrium value terminal condition when we replaced h by $F(y)$ in (6.30).

In order to examine the situation in which the constraints are binding, we look at the linear problem of maximizing

$$J[u] = \int_{t_0}^{b_1} [f_0\{t, y(t)\} + g_0\{t, y(t)\}u(t)] dt \quad (6.31)$$

subject to a state equation

$$y' = F(y) - u(t), \quad (6.32)$$

with $y(t_0) = y_0$, $y(t_1) = y_1$, and the control constraints

$$u_m \leq u(t) \leq u_M. \quad (6.33)$$

This problem is the special case $f_1(t, y) = F(y)$, $y_1(t, y) = -1$ of problem (6.5), (6.6) in Section 6.6, which was analyzed there by the use of line integrals and Green's theorem. For this problem, the Hamiltonian is

$$\begin{aligned} \mathcal{H}[t, y, u, \lambda] &= f_0(t, y) + g_0(t, y)u + \lambda(t)[F(y) - u] \\ &= f_0(t, y) + \lambda(t)F(y) + u[g_0(t, y) - \lambda(t)]. \end{aligned}$$

We let

$$\sigma(t) = g_0\{t, y(t)\} - \lambda(t). \quad (6.34)$$

Since $\mathcal{H}$ is linear in u , it is clear that in order to maximize $\mathcal{H}$ we must take $u = u_M$ if $\sigma(t) > 0$ and $u = u_m$ if $\sigma(t) < 0$. This is a bang-bang control and σ is called the *switching function*. When the switching function vanishes, $\mathcal{H}$ is independent of u . Thus binding constraints lead to bang-bang controls.

A singular case arises if $\sigma \equiv 0$ on an interval. To determine the corresponding *singular control*, we write $\sigma'(t) = 0$, or

$$\frac{\partial g_0}{\partial t} + \frac{\partial g_0}{\partial y} \frac{dy}{dt} - \frac{d\lambda}{dt} = 0.$$

From the state equation (6.32) and the adjoint equation (6.25) in the form

$$\frac{d\lambda}{dt} = -\frac{\partial}{\partial y}(f_0 + g_0u) - \lambda \frac{\partial}{\partial y}(F - u)$$

we have

$$\frac{\partial g_0}{\partial t} + \frac{\partial g_0}{\partial y}(F - u) + \frac{\partial f_0}{\partial y} + u \frac{\partial g_0}{\partial y} + \lambda \frac{\partial F}{\partial y} = 0. \quad \blacksquare$$

From (6.34) and the switching condition $\sigma(t) = 0$, this is equivalent to

$$-\frac{\partial g_0}{\partial t} = F \frac{\partial g_0}{\partial y} + \frac{\partial f_0}{\partial y} + g_0 \frac{\partial F}{\partial y}. \quad (6.35)$$

In Section 6.6 we used the state equation to write the integral

$$\int_{t_0}^{t_1} [f_0\{t, y(t)\} + g_0\{t, y(t)\}u(t)] dt$$

in the form

$$\int_{t_0}^{t_1} [G(t, y) + H(t, y)y'] dt,$$

and here we have

$$\begin{aligned} f_0(t, y) + g_0(t, y)u &= f_0(t, y) + g_0(t, y)[F(y) - y'] \\ &= f_0(t, y) + g_0(t, y)F(y) - y'g_0(t, y). \end{aligned}$$

Thus

$$\begin{aligned} G(t, y) &= f_0(t, y) + g_0(t, y)F(y), \\ H(t, y) &= g_0(t, y) \end{aligned}$$

and we have

$$\begin{aligned} \frac{\partial G}{\partial t} &= \frac{\partial f_0}{\partial y} + g_0 \frac{\partial F}{\partial y} + F \frac{\partial g_0}{\partial y}, \\ \frac{\partial H}{\partial t} &= -\frac{\partial g_0}{\partial t}. \end{aligned}$$

Now (6.35) is equivalent to

$$\frac{\partial H}{\partial t} = -\frac{\partial G}{\partial y},$$

which was the equation in Section 6.6 that was solved to obtain the function $y^*(t)$. The content of the maximum principle is that for this problem the optimal control is a combination of bang-bang and singular controls.

The optimization problem (6.19) and (6.20) formulated at the beginning of this section is not linear in h , and the method used in Section 6.6 cannot be applied. Here we are forced to use the maximum principle. The Hamiltonian is

$$\mathcal{H}(t, y, u, \lambda) = e^{-\delta t} R(u) + \lambda(t)[F(y) - u].$$

If the control constraints are not binding, the maximum equation is

$$\frac{\partial \mathcal{H}}{\partial u} = e^{-\delta t} R'(u) - \lambda(t) = 0,$$

or

$$\lambda(t) = e^{-\delta t} R'\{u(t)\}. \quad (6.36)$$

The adjoint equation is

$$\frac{dy}{dt} = -\frac{\partial \mathcal{H}}{\partial y} = -\lambda(t)F'(y) = -e^{-\delta t} R'\{u(t)\}F'(y).$$

From (6.36) we obtain

$$\frac{d\lambda}{dt} = e^{-\delta t} \left[R''(u) \frac{du}{dt} - \delta R'(u) \right],$$

and combination with the adjoint equation gives

$$R''(u) \frac{du}{dt} - \delta R'(u) = -R'(u)F'(y).$$

We now have a two-dimensional autonomous system for the response y and the control u , namely

$$y' = F(y) - u, \quad (6.37)$$

$$u' = \frac{R'(u)}{R''(u)} [\delta - F'(y)]. \quad (6.38)$$

It is reasonable to assume that the revenue function $R(u)$ satisfies $R(u) > 0$, $R'(u) \geq 0$, and $R''(u) < 0$.

The y -isocline of the system (6.37) is the curve $u = F(y)$, and the u -isocline is the vertical line $F'(y) = \delta$. There is a unique equilibrium (y^*, u^*) given by

$$u^* = F(y^*), \quad F'(y^*) = \delta,$$

and we may verify that this equilibrium is a saddle point, and that the separatrices and phase portrait are as shown in [Figure 6.23](#)

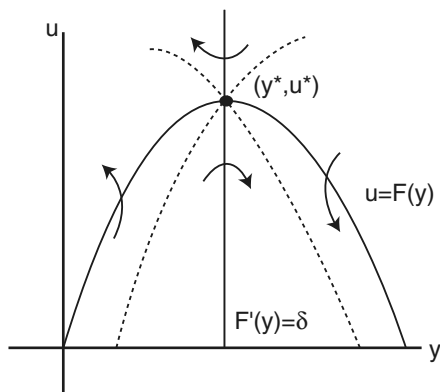


Fig. 6.23 Phase portrait with separatrices.

As in the linear problem of Section 6.5, there is an optimal equilibrium solution $y = y^*$, $u = u^*$. However, the optimal approach to equilibrium now is not a bang-bang control as in the linear case. The optimal trajectory must start on the line $y = y_0$ at time $t = 0$ and terminate on the line $y = y_T$ at time $t = T$. Since (y^*, u^*) is an equilibrium, the velocity along a trajectory near (y^*, u^*) is small. The closer an orbit is to the separatrices, the longer is the time taken to go from y_0 to y_T . Thus for small T , the trajectory will be far from the saddle point, and for large T , the trajectory will be close to the saddle point. For each point on $y = y_0$ there is a trajectory going

to the line $y = y_T$, and the optimal trajectory is the one taking the required time T (Figure 6.24).

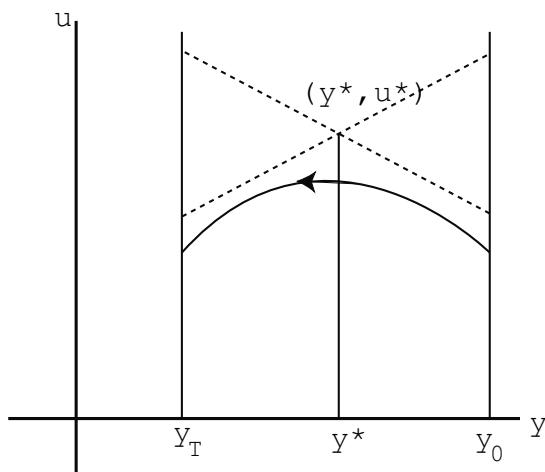


Fig. 6.24 Optimal trajectory.

As the time horizon T goes to infinity, the terminal condition drops out and the optimal trajectory is simply the part of the separatrix from $y = y_0$ to (y^*, u^*) . The optimal approach is more gradual than the bang-bang approach of the linear model. This reflects the dependence of prices on the harvest, which one would expect to cause more gradual changes in the optimal harvest rate.

6.8 Economic Interpretation of the Maximum Principle

We now give an interpretation of the maximum principle in economic terms. The computations that lead to this interpretation actually can be refined to give a proof of the maximum principle under the assumption that all functions involved are sufficiently smooth (twice differentiable). However, in practice the functions are not necessarily this smooth, as we have seen with bang-bang controls, and the general proof of the maximum principle is much more difficult. For a full proof, we refer the reader to the books of Pontryagin *et al.* (1962) and Lee & Markus (1968).

We consider the problem of maximizing

$$J[u] = \int_{t_0}^{t_1} g[s, y(s), u(s)] ds \quad (6.39)$$

subject to a state equation

$$y' = f(t, y, u) \quad (6.40)$$

with initial and terminal conditions

$$y(t_0) = y_0 \quad \text{and} \quad y(t_1) = y_1. \quad (6.41)$$

We view the terminal point (t_1, y_1) as fixed but treat the initial point (t_0, y_0) as variable, assuming that an optimal control $u^*(t)$ exists for all initial points (t_0, y_0) under consideration. For each initial point we define the function

$$w(t_0, y_0) = J[u^*] = \max J[u]. \quad (6.42)$$

This function $w(t_0, y_0)$ in economic problems represents the present value of the stock level y_0 at time t_0 under the assumption that an optimal exploitation policy will be used for $t_0 \leq t \leq t_1$. Thus $w(t_0, y_0)$ is the value of the capital asset y_0 at time t_0 with a given time horizon t_1 . Because the optimal policy must be optimal at each point of the interval $t_0 \leq t \leq t_1$, we must have

$$w(t_0, y_0) = \int_{t_0}^t g[s, y^*(s), u^*(s)] ds + u[t, y^*(t)] \quad (6.43)$$

for every t . Differentiation of (6.42) with respect to t gives

$$\begin{aligned} \frac{d}{dt} w[t, y^*(t)] &= \frac{\partial w}{\partial t} [t, y^*(t)] + \frac{\partial w}{\partial y} [t, y^*(t)] \frac{dy}{dt} \\ &= \frac{\partial w}{\partial t} [t, y^*(t)] + \frac{\partial w}{\partial y} [t, y^*(t)] f[t, y^*(t), u^*(t)] \\ &= -g[t, y^*(t), u^*(t)]. \end{aligned} \quad (6.44)$$

If we define

$$\lambda(t) = \frac{\partial w}{\partial y} [t, y^*(t)] \quad (6.45)$$

and the Hamiltonian

$$\mathcal{H}[t, y, u, \lambda] = g(t, y, u) + \lambda f(t, y, u), \quad (6.46)$$

the maximum principle implies that

$$\mathcal{H}[t, y^*(t), u^*(t), \lambda(t)] = \max_u \mathcal{H}[t, y, u, \lambda]. \quad (6.47)$$

The function $G(t, y, u)$ defined by

$$G(t, y, u) = \frac{\partial w}{\partial t} (t, y) + \frac{\partial w}{\partial y} (t, y) f(t, y, u) + g(t, y, u) \quad (6.48)$$

has the property that

$$G(t, y^*(t), u^*(t)) = 0 \quad (6.49)$$

because of (6.44). It is possible to show, by an argument that involves considering the effect of varying the control u in (6.43), that

$$G(t, y, u) \leq 0 \quad (6.50)$$

for all (t, y, u) . From (6.48) and (6.49) we see that $G(t, y, u)$ is maximized by the choice $y = y^*(t)$, $u = u^*(t)$. In particular, if we fix $y = y^*(t)$, then $G(t, y^*(t), u)$ is maximized by the choice $u = u^*(t)$, and if we fix $u = u^*(t)$ then $G(t, y, u^*(t))$ is maximized by the choice $y = y^*(t)$. As an alternative to this maximization argument, we could use the maximum principle to show that $G(t, y, u)$ is maximized by the choice $u = u^*(t)$, since

$$\begin{aligned} \mathcal{H}[t, y^*(t), u, \lambda] &= g(t, y^*(t), u) + \lambda f(t, y^*(t), u) \\ &= g(t, y^*(t), u) + \frac{\partial w}{\partial y}(t, y^*(t)) f(t, y^*(t), u) \\ &= G(t, y^*(t), u) - \frac{\partial w}{\partial t}(t, y^*(t)) \end{aligned}$$

by (6.45), (6.46), and (6.48); maximization of $G(t, y^*(t), u)$ is equivalent to maximization of $\mathcal{H}[t, y^*(t), u, \lambda]$.

Now we fix $u = u^*(t)$ and use the fact that $G(t, y^*(t), u)$ attains its maximum value when $y = y^*(t)$. If G is differentiable, we must have

$$\frac{\partial G}{\partial y}(t, y^*(t), u^*(t)) = 0. \quad (6.51)$$

Differentiation of (6.48) shows that (6.51) implies

$$\frac{\partial^2 w}{\partial y^2} f + \frac{\partial w}{\partial y} \frac{\partial f}{\partial y} + \frac{\partial^2 w}{\partial y \partial t} + \frac{\partial g}{\partial y} = 0, \quad (6.52)$$

with all functions being evaluated for $y = y^*(t)$, $u = u^*(t)$. Differentiation of (6.45) gives

$$\frac{d\lambda}{dt} = \frac{\partial^2 w}{\partial y^2} \frac{dy}{dt} f + \frac{\partial^2 w}{\partial t \partial y} = \frac{\partial^2 w}{\partial y^2} f + \frac{\partial^2 w}{\partial t \partial y}. \quad (6.53)$$

Combination of (6.52) and (6.53) gives

$$\frac{d\lambda}{dt} + \frac{\partial w}{\partial y} \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y} = 0,$$

or

$$\frac{d\lambda}{dt} = -\lambda \frac{\partial f}{\partial y} - \frac{\partial g}{\partial y} = -\frac{\partial \mathcal{H}}{\partial y}, \quad (6.54)$$

which is the adjoint equation.

The above calculation shows that we may interpret the adjoint variable λ in the maximum principle as being given by the expression (6.45). Since $w(t, y)$ repre-

sents the value of the capital asset y at time t , the adjoint variable λ represents the *marginal value* of this capital asset at time t . Reduction of the capital by one unit would reduce the value by approximately λ ; for this reason $\lambda(t)$ is called the *shadow price* of the asset.

The Hamiltonian is the sum of two terms, each of which can be interpreted as a flow of values. The term $g(t, y, u)$ is the flow of revenues obtained from resource harvesting with the objective functional (6.39). The quantity $f(t, y, u)$ is the flow of investment in capital,

$$\frac{dy}{dt} = f(t, y, u).$$

The value of this investment flow is the flow multiplied by the shadow price $\lambda(t)$. Thus, the Hamiltonian represents the rate of increase of total assets, both accumulated dividends and capital assets. The maximum principle says that an optimal control must maximize the rate of increase of total assets, but we must know the shadow price in order to determine the optimal control. In effect, the maximum principle reduces the optimal control problem to the question of determining the shadow price. Unfortunately, the analytical determination of the shadow price in practice is a formidable problem, and it is often impossible to obtain an exact expression for the shadow price.

The adjoint equation (6.51) says that the rate of depreciation of capital $-\frac{d\lambda}{dt}$ consists of two parts, the marginal flow to accumulated dividends and the marginal flow to capital assets. Along the optimal path, this rate of depreciation must be equal to the sum of these marginal flows, $\frac{\partial g}{\partial y}$ and $\lambda \frac{\partial f}{\partial y}$, respectively. The economic literature usually speaks of a *current shadow price* $\tilde{\lambda}(t) = e^{\delta t} \lambda(t)$, rather than the discounted shadow price $\lambda(t)$ and of the *current-value* Hamiltonian

$$\tilde{\mathcal{H}} = e^{\delta t} \mathcal{H}.$$

In these terms the adjoint equation $\frac{d\lambda}{dt} = -\frac{\partial \mathcal{H}}{\partial y}$ becomes $\frac{d\tilde{\lambda}}{dt} = \delta \tilde{\lambda} - \frac{\partial \tilde{\mathcal{H}}}{\partial y}$.

6.9 Project: A Harvesting Model

Consider a predator–prey system described by the model

$$\begin{aligned} y' &= ry \left(1 - \frac{y}{K}\right) - \frac{ayz}{y+A} \\ z' &= sz \left(1 - \frac{z}{by}\right), \end{aligned}$$

where r, s, a, b, K are positive constants. Suppose the predator (z) species is harvested at a constant rate H .

1. Find the coexistence equilibrium of the unharvested system.

2. Show that under harvesting the equilibrium lies on a parabola opening downwards and passing through the points $(-A, 0)$, $(K, 0)$.
3. Show that for $H > 0$, the equilibrium (y, z) satisfies $z < by$.
4. Is it possible to drive the predator population to zero by harvesting? If so, what is the smallest harvest rate that achieves this?

6.10 Project: Harvesting of Two Species

Consider a predator–prey system given by the pair of equations

$$\begin{aligned}x' &= xf(x, y), \\ y' &= yg(x),\end{aligned}\tag{6.55}$$

with

$$f_x(x, y) < 0, \quad g_x(x, y) > 0, \quad f(K, 0) = 0, \quad g(J, 0) = 0.$$

Assume constant-yield harvesting of both species, to give a model

$$\begin{aligned}x' &= xf(x, y) - H_x, \\ y' &= yg(x) - H_y.\end{aligned}\tag{6.56}$$

Carry out as much as possible of the analogue to the theory of Section 6.2 for harvesting of only the predator species. Instead of a single critical harvest, you should expect to find a curve

$$h[H_x, H_y] = 0$$

of critical harvest rates.